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AI-Powered Research Paper Summarization and Question Answering System

Interim Project Demo by

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ABSTRACT

Problem: Students and researchers spend a lot of time reading many research papers to find useful information for their projects. Existing platforms like Google Scholar and Semantic Scholar only help with searching papers, but they do not give short summaries or answer specific questions from the papers. This makes it hard to quickly understand the key points, limitations, and future work of previous studies.

Methodology: This project builds an AI-based system that can find, summarize, and answer questions about research papers. First, the system collects papers from online sources using web scraping. Then, it extracts text from PDFs and paper abstracts. Next, a transformer-based NLP model T5-base is used for both summarization and question answering tasks. The model summarizes the research papers by generating concise summaries of their content, and also answers user questions such as "What are the limitations?" or "What are the future works?" by understanding the paper's context. A simple user interface is built using Streamlit or React, and the backend uses Python with Flask or FastAPI to process model requests and display the results to users.

Subject Descriptors:

Information systems -> Information retrieval -> Search engines -> Document indexing

Computing methodologies -> Natural language processing -> Summarization -> Question answering

Keywords: Research summary, question answering, NLP, AI assistant, academic papers

DECLARATION

I so certify that all the study components are original to me, have not been submitted before, and will not be submitted to any other degree program.

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List of Abbreviations

Abbreviation	Definition
NLP	Natural Language Processing
QA	Question Answering
BART	Bidirectional and Auto-Regressive Transformer
API	Application Programming Interface
UI	User Interface
AI	Artificial Intelligence
SRS	Software Requirement Specification
SLEP	Social, Legal, Ethical, and Professional

Chapter 01: Introduction

1.1 Chapter Overview

In today's world, a huge number of research papers are published every year, making it difficult for students and researchers to find and understand the information they need quickly. Websites like Google Scholar and Semantic Scholar help people find papers, but they do not explain or summarize the content. This often causes information overload and wastes time when trying to identify the key ideas, results, and limitations of studies. Recent progress in AI and NLP has made it possible to automatically summarize and answer questions from research papers. Models such as BART, T5, and Longformer can read and summarize long documents and even generate human-like answers to user questions. However, many current systems still face problems like giving incomplete or incorrect answers and struggling with long or complex papers. To solve these issues, this project aims to build an AI-Based Research Summary and Question Answering Application that can search, summarize, and answer questions about research papers in one platform. The system will use transformer-based AI models to understand full papers and give accurate summaries and clear answers. This tool is designed to help students and researchers save time, understand studies easily, and improve the process of doing literature reviews.

1.2 Problem Background

1.2.1 Information Overload in Research

A massive number of research papers are published daily across various scientific fields. Platforms like arXiv, IEEE Xplore, and ScienceDirect make it easy to access these papers, but the increasing volume of publications has caused a significant challenge information overload (Berlant et al., 2022). Students and researchers working on academic projects often struggle to go through hundreds of papers to understand existing studies, methods and findings related to their topics (Ali et al., 2024).

1.2.2 Time Consuming and Inefficient Manual Reading

Reading and analyzing multiple research papers manually takes a lot of time and effort. While research papers often include abstracts, these don't provide a complete understanding of the paper's key contributions, limitations and future work (Kryscinski et al., 2020). It

becomes difficult for students and researchers to identify the most relevant papers for their studies or projects (Nassiri et al., 2023).

1.3 Problem Definition

1.3.1 Challenge of Reviewing Numerous Research Papers

Students and researchers working on academic projects are required to review a large number of research papers to understand existing studies, research methods and findings within their area of interest. However, the continuous exponential growth in the number of papers published across platforms such as arXiv, IEEE Xplore and ScienceDirect makes it nearly impossible to manually read and analyze all relevant studies (Berlant et al., 2022). This leads to information overload, where users struggle to filter out the most important content from an ever growing collection of academic literature, making the research process inefficient and overwhelming for beginners (Ali et al., 2024).

1.3.2 Limitations of Current Digital Libraries and Academic Search Engines

While digital libraries and academic search engines help users locate relevant papers, they do not assist in the understanding or summarization of the papers themselves (Nassiri et al., 2023). Existing systems primarily focus on paper retrieval and citation analysis, but fail to provide deep comprehension of the research content. Most systems that attempt summarization either use extractive methods, which simply copy sentences from the text, or they lack factual accuracy and coherence (Kryscinski et al., 2020).

1.3.3 Lack of Interactive Question Answering Systems

Very few existing systems include an interactive QA feature that allows users to ask specific questions about a paper and receive meaningful responses. For example, users may want to ask about the paper's limitations, methodology, or future directions (Mallick et al., 2023). Without this feature, users are left with summaries that don’t directly address specific queries they might have about the paper.

Problem Statement

Manually reviewing large volumes of research papers is time consuming and prone to information overload. This project aims to develop an AI-based system using the BART-Base

model to automatically summarize papers and answer related questions, helping users quickly understand academic literature.

1.4 Research Motivation

The rapid growth of scientific research has led to an overwhelming increase in published papers, making it difficult for researchers to stay updated. Manually reviewing and analyzing papers requires significant time and effort, especially for early-stage researchers. Advances in AI and NLP have made it possible to develop models that can summarizeres each papers efficiently, such as BART, which excels at generating concise summaries while retaining important details. Modern research also demands interactive and personalized tools for querying specific aspects of papers, turning literature review into an active exploration. This project aims to bridge the gap between the growing number of publications and the limited capacity to process them, by developing an AI-based summarization system with BART and an interactive question-answering system with DistilBERT, ultimately enhancing research accessibility and promoting efficient knowledge discovery.

1.5 Existing Work

Citation	**Summary**	**Limitation**	**Contribution**
Sundaram Berleant (2022)	Conducted a systematic review on automating Systematic Literature Reviews (SLR) using NLP, Text Mining, and AI techniques. Analyzed 29 published studies focusing on automation of study selection and data	Still relies on partial manual work for data extraction and synthesis; limited automation in key SLR areas.	Provided a detailed overview of how NLP and ML methods can automate literature review steps, highlighting progress in SLR automation.

	extraction.		
N. Fateh Ali et al. (2024)	Developed an automated literature review system using NLP and LLM-based Retrieval-Augmented Generation (RAG). Compared models like spaCy, T5, and GPT-3.5-turbo on the SciTLDR dataset for summarization.	Accuracy limited by dataset quality and model context length; current models cannot fully capture deep research insights.	Proposed a comparative evaluation of multiple summarization methods and introduced a GUI-based automated summarization tool.
H. Jin et al. (2024)	Conducted a comprehensive survey on automatic text summarization methods using LLMs like GPT, across domains such as research, news, and legal. Proposed a new retrieval algorithm for collecting papers.	LLMs face hallucination, lack of explainability, and domain inefficiency; prompt design limitations.	Offered an updated survey on LLM-based summarization, suggesting improvements in prompt design and fine-tuning for domain-specific summarization.
Veena R., D. Ramesh, and Hanumantappa M. (2023)	Presented a systematic literature review on automatic text summarization using AI. Compared	Limited number of papers reviewed; lacks practical implementation and quantitative	Summarized how AI has evolved summarization methods and identified the need

	extractive and abstractive summarization techniques and their evolution.	evaluation.	for real-time abstractive models and hybrid approaches.
Zaman F. et al. (2024)	Proposed SATS, a transformer-based model that performs simplification-aware summarization of scientific documents.	Focused more on readability and simplification, which can reduce the depth of technical content.	Summarized how AI has evolved summarization methods and identified the need for real-time abstractive models and hybrid approaches.

1.6 Research Gap

1.6.1 Limitations of Existing Summarization Models

- **Current Models:** Models like T5, PEGASUS, and GPT-based systems have made significant progress in generating coherent and concise summaries of scientific texts (Lewis et al., 2020; Daraghmi et al., 2024).
- **Limitations:** Despite strong performance on benchmark datasets, these models face limitations when applied to full length academic papers:
    - **Dataset Quality:** Many models are trained on general purpose datasets, such as CNN/DailyMail, which are not suitable for the technical language of academic writing (Kryscinski et al., 2020; Nassiri et al., 2023).
    - **Context Handling:** Standard Transformer models are limited by their token length, making it difficult to summarize long documents like research papers or dissertations (Beltagy et al., 2020).

1.6.2 Newer Models and Computational Costs

- **Challenges with Newer Models:** Models like Longformer and LED attempt to overcome these limitations but require extensive computational resources and fine-tuning for accurate results (Rehman et al., 2023).
- **Hallucination and Low Explainability:** Large Language Models (LLMs) like GPT-3 and GPT-4 still suffer from issues such as generating factually incorrect content (hallucination), low explainability, and high computational costs, making them difficult to deploy in resource-limited environments (Nassiri et al., 2023; He et al., 2024).

1.6.3 Lack of Domain Specific Models

- **Insufficient Adaptation:** Many summarization models are not fine-tuned for the academic domain, where technical vocabulary, formulas, and complex sentence structures are common (Yue et al., 2021).
- **Misinterpretation of Context:** Without domain adaptation, general purpose summarization models often miss critical contextual meanings or distort key findings in academic papers.

1.6.4 Gap in Interactive Question Answering Systems

- **Current Focus:** Existing systems primarily focus on summarization and overlook the interactive QA component, which is essential for a deeper understanding and exploration of research content (Mallick et al., 2023).
- **Lack of Engagement:** Without QA features, users cannot interact with the summarized information in a meaningful and query driven way, limiting their ability to gain precise insights.

Proposed Solution: AI-Based Summarization and QA System

- **BART-Base Model:** This project aims to bridge the gaps using the BART-Base model, which balances efficiency, accuracy, and interpretability for abstractive summarization (Lewis et al., 2020).
- **Fine-Tuning for Academic Content:** Unlike purely generative LLMs, BART can be fine-tuned for scientific text summarization, improving factual consistency and domain specific understanding (Rehman et al., 2023).
- **QA Component:** In addition to summarization, this project integrates a QA component, allowing users to interactively query research papers and extract

meaningful insights, such as methodologies, limitations, and future directions (Mallick et al., 2023; He et al., 2024).

Transforming Passive Consumption to Active Exploration

- **Dual Functionality:** By combining summarization and interactive question answering, the proposed system transforms passive information consumption into active knowledge exploration.
- **Contribution to Academic Research:** This dual function framework addresses the need for domain specific, efficient, and interactive AI systems for academic research, making it more accessible, intelligent, and user centered (Pearce et al., 2021).

1.7 Contribution to Body of Knowledge

Problem Domain Contribution

- **Addressing the Challenge of Information Overload**

This research significantly contributes to tackling the growing challenge of efficiently understanding and extracting meaningful insights from the rapidly expanding volume of academic literature. Every day, thousands of new research papers are added to repositories such as arXiv, IEEE Xplore, and ScienceDirect, making it increasingly difficult for researchers and students to stay updated with relevant studies (Pearce et al., 2021). Traditional manual methods of reviewing and analyzing these papers are time-consuming and cognitively demanding, often leading to information overload and inefficiency (Sundaram & Berleant, 2022).

- **Use of BART-Base Model for Summarization**

To overcome this limitation, the proposed system employs the BART-Base model, a Transformer based architecture designed for abstractive text summarization. Unlike extractive methods, which simply select sentences from the original text, BART's abstractive mechanism rephrases content in a human-like manner. This preserves the contextual meaning while significantly reducing reading time (Lewis et al., 2020). As a result, researchers can quickly grasp the main objectives, methodologies, and results of studies without having to read each paper in full.

- **Integration of Question Answering (QA) Module**

In addition to summarization, the system includes a Question Answering (QA) module that enhances user interactivity. This allows users to query specific aspects of the summarized research, such as the study's limitations, key methods, or future work (Mallick et al., 2023; He et al., 2024). By integrating both summarization and QA capabilities, the system transforms the traditional passive literature review process into an interactive knowledge retrieval experience, improving accessibility and research efficiency.

- **Contribution to Students, Educators, and Early-Stage Researchers**

This contribution is especially valuable for students, educators, and early-stage researchers, as it democratizes access to complex academic information. The proposed system enables users to extract domain relevant insights more efficiently, supporting informed decision making during the research planning phase. Additionally, it reduces the time spent in the initial stages of literature review (Daraghmi et al., 2024). By making research papers more accessible, the system helps bridge the knowledge gap for those with limited experience in academic research.

- **Impact on Research Productivity and Intelligent Automation**

Overall, this system contributes to improving research productivity by helping researchers quickly navigate academic literature. It also promotes intelligent automation in academic knowledge management, enabling more effective use of time and resources during the research process. This system is not only a tool for summarization but also a step toward more efficient academic workflows.

Research Domain Contribution

- **Advancing NLP and Deep Learning for Domain Specific Summarization**

From a research and technical standpoint, this project makes significant advancements in Natural Language Processing (NLP) and Deep Learning by fine-tuning the BART-Base Transformer model for domain specific scientific text summarization and question answering. These are two complex NLP tasks that require both linguistic understanding and factual consistency. While models like T5, PEGASUS, and GPT-3 have achieved high performance

on general summarization tasks, they often struggle with domain adaptation and factual accuracy when applied to research oriented datasets (Yue et al., 2021; Nassiri et al., 2023).

- **Fine-Tuning BART for Scientific Texts**

This study demonstrates how fine-tuning BART on scientific corpora enhances its ability to capture technical terminology, structured argumentation, and domain specific nuances found in research papers. This adaptation helps improve the model's ability to process and summarize technical content more accurately, making it more suitable for academic use (Rehman et al., 2023).

- **Contribution to Multi-Task Learning**

The project also contributes to the field of multi-task learning by exploring how summarization and question answering (QA) can be effectively integrated into a single unified framework. Such integration not only improves the model's contextual comprehension but also provides a foundation for future NLP systems capable of supporting multiple academic information processing tasks simultaneously (He et al., 2024).

- **Lightweight Models and Computational Efficiency**

Additionally, the study provides insights into how lightweight Transformer models like BART can achieve strong performance with lower computational overhead compared to larger models such as GPT-4 or PaLM. This helps promote energy efficient and accessible AI solutions for research applications, making it feasible for researchers with limited computational resources to use these models effectively (Nassiri et al., 2023).

- **Contributions to the Body of Scientific Knowledge**

This project contributes to the body of scientific knowledge by advancing the understanding of how Transformer-based summarization and QA models can be optimized for research intensive domains. It lays the groundwork for future studies exploring explainable and factually consistent AI systems that can support academic research automation (Sundaram & Berleant, 2022).

- **Addressing Limitations in Summarization, Interactivity, and Scalability**

By addressing current limitations in summarization accuracy, interactivity, and scalability, this research extends the boundaries of NLP applications in the academic field. It enables faster, smarter, and more reliable information discovery, providing tools for researchers to access and interpret academic knowledge more efficiently.

1.8 Research Challenges

- **Balancing Summary Accuracy and Contextual Completeness**

One of the main challenges of this project is maintaining a balance between summary accuracy and contextual completeness. The goal is to ensure that the generated summaries accurately capture the essential meaning of research papers without losing important technical details. Achieving this balance is crucial to ensure that the summaries are both concise and informative.

- **Fine-Tuning BART for Domain Specific Language**

Another significant challenge lies in fine-tuning the BART-Base model for highly domain specific academic language. Academic writing often includes complex vocabulary and sentence structures that differ greatly from general text, making it difficult to train models effectively for this type of content. The need to adapt the model to handle this specialized language is a key obstacle.

- **Handling Large, Unstructured Datasets**

The project also faces difficulties in handling large, unstructured datasets from platforms like arXiv. Issues include:

- **Data cleaning:** Ensuring that the data is free from errors and inconsistencies.
- **Text extraction:** Extracting text from PDF formats, which can be messy and difficult to parse.
- **Data organization:** Structuring the data in a way that makes it suitable for efficient model training.

- **Ensuring Accurate Question Answering**

Ensuring that the question answering system retrieves precise, factually correct responses from the summarized text is another challenge. This requires advanced model calibration and

careful evaluation to ensure that the answers are accurate and relevant to the user's queries. Fine-tuning the model for this task is complex, especially when dealing with abstracted and shortened content.

- **Computational and Performance Challenges**

Finally, the project faces computational and performance challenges. Transformer based models are resource intensive, requiring significant computational power. To run efficiently on limited hardware, the models must be optimized without compromising the quality of the results. Ensuring that the system works efficiently while maintaining high quality output is an ongoing challenge.

1.9 Research Questions

This research aims to address the following key questions within the scope of automated research paper understanding and summarization:

1. How can transformer-based models such as BART-base be effectively fine-tuned to generate concise and contextually accurate summaries of research papers?
2. What preprocessing and text-cleaning techniques can improve the quality and relevance of summaries produced from large academic datasets like arXiv?
3. How can the integration of summarization and question answering systems enhance the accessibility and usability of complex research content for users?
4. What are the most effective evaluation metrics (e.g., ROUGE, BLEU, BERTScore) to measure the performance and factual accuracy of the generated summaries and responses?

1.10 Research Aim

Primary Aim of the Research

The primary aim of this research is to develop an AI-based research summarization and question answering system that can automatically retrieve, analyze, and generate concise summaries of research papers using the BART-Base Transformer model. This system is designed to help students, educators, and researchers efficiently understand large volumes of academic content by providing short, coherent, and contextually meaningful summaries that highlight the core ideas, methodologies, results, and limitations of research papers.

The Growing Challenge in Academic Research

In today's academic environment, the vast amount of information available across repositories such as arXiv, IEEE Xplore, and ScienceDirect makes it increasingly difficult for individuals to manually read and interpret every relevant study (Pearce et al., 2021; Sundaram & Berleant, 2022). As a result, the literature review process becomes cumbersome and time consuming, highlighting the need for automated tools that can assist in streamlining this task.

Automating the Literature Review Process

To address this challenge, the project aims to build a tool that automates the literature review process using state-of-the-art Natural Language Processing (NLP) and Deep Learning techniques. By leveraging the BART-Base model (Lewis et al., 2020), which excels in abstractive text summarization, the system can understand complex scientific language and produce human-like summaries that retain the essential meaning of the source content.

Integration of Question Answering Capability

In addition to summarization, the system will support an interactive QA, feature, allowing users to ask specific, targeted questions related to the paper. For example, users can query about the research problem, experimental methods, findings, limitations, or future work. This feature transforms the application from a passive summarization tool into an intelligent research assistant that actively supports knowledge discovery (Mallick et al., 2023; He et al., 2024).

Reducing Time and Cognitive Load in Research

Ultimately, the goal of this research is to reduce the time, effort, and cognitive load required for literature review and academic exploration. By combining automatic summarization and question answering within a single framework, the project aims to make the research process more efficient, accessible, and interactive, empowering students and researchers to quickly identify relevant studies, understand their contributions, and build upon existing knowledge.

1.11 Research Objectives

Objectives	Research	Description	LOs Mapped	RQ Mapped

	Objective			
Problem Identification	**R01:**To identify the research gap in the area of research paper summarization and question answering using transformer-based models.	Identify the limitations of current summarization and QA systems, such as low contextual accuracy and lack of interactivity.	**L02**	**RQ1, RQ3**
Literature Review	**R02:** To carry out an in-depth literature survey on research paper summarization using NLP techniques.	Review studies related to text summarization and question answering to understand existing methodologies and their limitations.	**L01, L04**	**RQ1, RQ2**
	R03:To conduct a literature review to choose the best available NLP-based	Compare different transformer-based models (e.g., T5, BART, GPT) and	**L01, L04**	**RQ1, RQ2**

	models for improving summarization and answering performance.	justify the selection of BART-Base for this project.		
Requirement Elicitation	**R04:**To identify the stakeholders and user requirements for developing a summarization and QA application.	Define user expectations, input/output formats, dataset sources, and model requirements for system development.	**L03**	**RQ4**
System Design	**R05:** To design the architecture of the summarization and QA system.	Design the system's architecture, including the integration of the BART-Base model for summarization and the DistilBERT model for question answering.	**L06**	**RQ4**

Implementation	**R06:** To implement the summarization and QA system using the selected models (BART-Base and DistilBERT).	Integrate BART-Base for summarization and DistilBERT for question answering into a working system.	**L05**	**RQ2, RQ3**
Testing	**R07:** To evaluate the performance of the summarization and QA system.	Perform testing and benchmarking of the system's summarization accuracy and question-answering performance.	**L07**	**RQ3, RQ4**
Documentation	**R08:** To document the development process and final results.	Provide detailed documentation on the system's design, implementation, and evaluation results.	**L08**	**RQ4**

Publication	**R09:** To publish the literature survey and final research paper with the results.	Publish the literature survey outlining key findings and the final paper detailing the system, its functionality, and evaluation.	**L09**	**RQ4**

1.12 Project Scope

1.12.1 In Scope

- The system will work with research papers written in English only.
- The system will extract key sections of research papers like abstracts, introductions, methodologies, results, discussions and conclusions.
- The system will use the BART-Base Transformer model for abstractive text summarization and DistilBERT for question answering tasks .
- The system will focus on technical and scientific domains such as computer science, engineering, and healthcare.
- The system will evaluate performance using ROUGE(for summarization) and BLEU (for response accuracy).
- The system will generate high-quality summaries and provide answers to targeted questions based on research papers.

1.12.2 Out Scope

- The system will not process research papers written in languages other than English.
- The system will not analyze multimodal data such as images, charts, figures and tables.
- The project will not fine-tune or integrate other large-scale transformer models beyond BART-Base.

- The system will not perform real-time web scraping or automatic collection of research papers.
- The system will not evaluate or rank the credibility, impact or quality of research papers.
- The summaries and QA responses will not undergo human validation or expert review in this phase.

1.13 Hardware/Software Requirements

Hardware Requirements

Component	**Description**
Processor (CPU)	A computer system with a modern multi-core processor suitable for machine learning development.
Memory (RAM)	Sufficient memory to handle data preprocessing and model training tasks smoothly.
Storage	Adequate disk space to store datasets, model checkpoints, and experimental results.
Graphics Processing Unit (GPU)	A GPU-enabled system to support faster deep learning model training and inference.
Display	Standard display monitor for development and visualization of results.

Software Requirements

Software / Tool	**Purpose**
Python (Version 3.x)	Core programming language for implementing the AI model and data preprocessing.
PyTorch / TensorFlow	Deep learning framework for training and fine-tuning the BART-Base model.

Jupyter Notebook / Google Colab	Interactive environment for experimentation, code testing, and model analysis.
Scikit-learn	For preprocessing, performance evaluation, and utility functions.
Pandas & NumPy	For dataset handling and numerical computations.
Matplotlib / Seaborn	For result visualization and performance graph plotting.
Google Docs / Microsoft Word	For documentation and report preparation.
Git / GitHub	For version control and project collaboration.

1.14 Chapter Summary

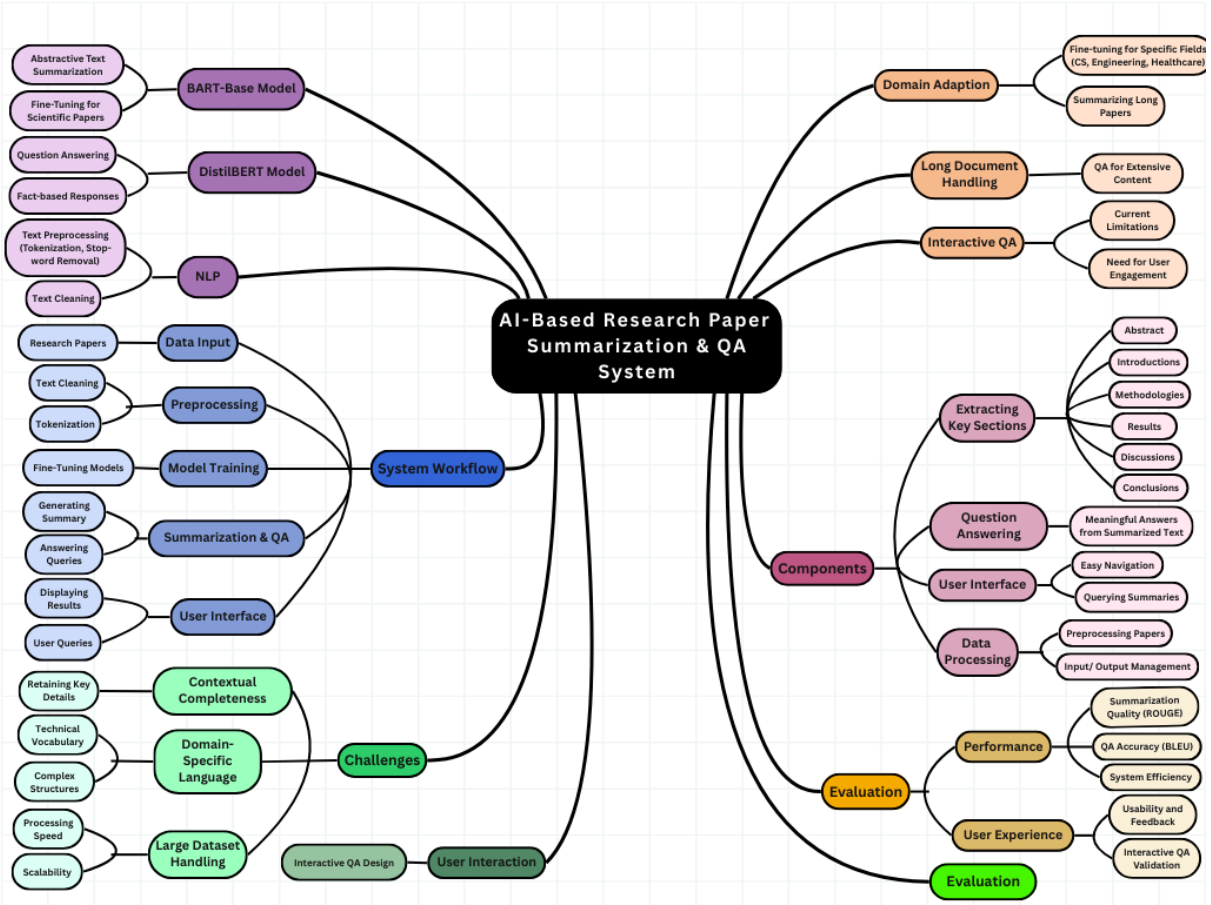
This chapter provided a comprehensive introduction and context for the research, establishing the foundation for the study on developing an AI-based system for automated research paper summarization using the BART-Base model and question answering using the DistilBERT model. It began by presenting the background and motivation, emphasizing the challenges faced by students and researchers in managing the ever-growing volume of scientific literature and the time consuming nature of manual literature reviews. The problem definition and statement highlighted the limitations of existing tools, particularly the lack of interactive summarization and question-answering capabilities, which motivated the need for a more efficient and intelligent system. Relevant studies in natural language processing, deep learning, and transformer-based models were reviewed, identifying key gaps such as limited context handling, domain-specific adaptation, and minimal integration of interactive question answering features. The chapter also discussed the expected contributions of the project, including the generation of concise, accurate summaries, improved contextual understanding, and the ability to respond to targeted queries, thereby enhancing research accessibility and efficiency. Additionally, research challenges, objectives, and scope were outlined, along with the required hardware and software environments, providing a clear framework and direction for the subsequent chapters that will focus on system design, implementation, evaluation, and potential future enhancements.

Chapter 02: Literature Review

2.1 Chapter Overview

This literature review focuses on abstractive summarization of English language scientific papers, specifically using transformer-based models like BART, T5 and PEGASUS which excel at generating human-like summaries by capturing contextual relationships rather than merely selecting sentences. The goal is to identify gaps in current systems, particularly in domain adaptation, long-document handling, factual consistency and interactive question answering and to motivate the use of BART for improvements in this project.

2.2 Concept Map



2.3 Problem Domain

The problem domain of this research lies within Natural Language Processing (NLP), specifically in the area of automatic research paper summarization. With the rapid growth of scientific publications, researchers and students face difficulties reading and understanding

large numbers of papers to find relevant information. Manual summarization is time consuming and often leads to missing key points.

To solve this, AI-based summarization systems are designed to automatically generate short, meaningful summaries from long research papers. These systems use machine learning and transformer based NLP models to understand context, identify important sentences, and create concise summaries that retain the core ideas. The use of models like BART allows more accurate and human-like summaries compared to older extractive methods.

2.3.1 Overview of Continuous Authentication

Abstractive summarization refers to the process of automatically rewriting long documents into concise, human-like summaries that preserve the core concepts, methodologies, and findings of the original text rather than simply selecting and re-ordering sentences (i.e., extractive summarization). Transformer-based sequence-to-sequence architectures such as BART (Bidirectional and Autoregressive Transformer) and PEGASUS have become widely adopted because they model long-range dependencies, capture contextual relationships and generate fluent summaries (Goodwin et al., 2020; “Flight of the PEGASUS?”). For example, studies have shown that BART, T5 and PEGASUS variants provide significant improvements in ROUGE/ BLEU metrics on benchmark summarization tasks (Munyao et al., 2024). In the context of research paper summarization, abstractive methods are particularly relevant because research texts often require paraphrasing rather than simply extracting key sentences, due to dense technical language, long documents, and the need to preserve the argument flow rather than isolated statements.

2.3.2 Importance of Behavioral Biometrics

Automated summarization systems play an increasingly important role for students, researchers and professionals who must sift through large volumes of academic literature. By generating concise summaries of many research papers, such tools save time, reduce manual reading effort and increase the ability to identify research trends, method gaps, and future directions across large document collections (e.g., arXiv, Google Scholar). Such summarization systems support more efficient literature reviews, enable faster scanning of relevant works, and improve decision making about which papers warrant deeper reading. In academic settings, where the volume of publications is increasing rapidly, this functionality can significantly decrease cognitive load and speed up the research cycle.

2.3.3 Application Areas

The proposed AI-based research summary and question answering system has applicability across several real world domains where quick understanding of research content is essential:

2.2.3.1 Security Sensitive Applications

In contexts such as internal institutional research, government funded projects or proprietary research labs where sensitive or unpublished documents are involved, the system must ensure safe handling of research data. Automated summarization and QA can be adapted for internal databases, reducing the risk of data leaks or unauthorized access and providing an efficient way to digest internal research without exposing full details externally.

2.2.3.2 User Privacy and Convenience

For the user side, the application should prioritise privacy by supporting local processing of uploaded papers or secure channels for cloud inference. It offers non-technical users (students, early stage researchers) a convenient interface (e.g., Streamlit or React) to obtain research summaries and answers without reading entire papers manually. This enhances accessibility, promotes personalised information retrieval, and thereby improves research productivity and inclusion for users with less domain expertise.

2.3.4 Challenges in Continuous Authentication

2.3.4.1 Maintaining Quality and Coherence

One of the major challenges in applying abstractive summarisation to scientific literature is avoiding hallucination where the model generates grammatically plausible summaries that are factually incorrect or omit critical technical details (e.g. studies on factuality in summarization). Ensuring that the summaries remain both coherent and factually accurate is a key bottleneck (Cao & Wang, 2021).

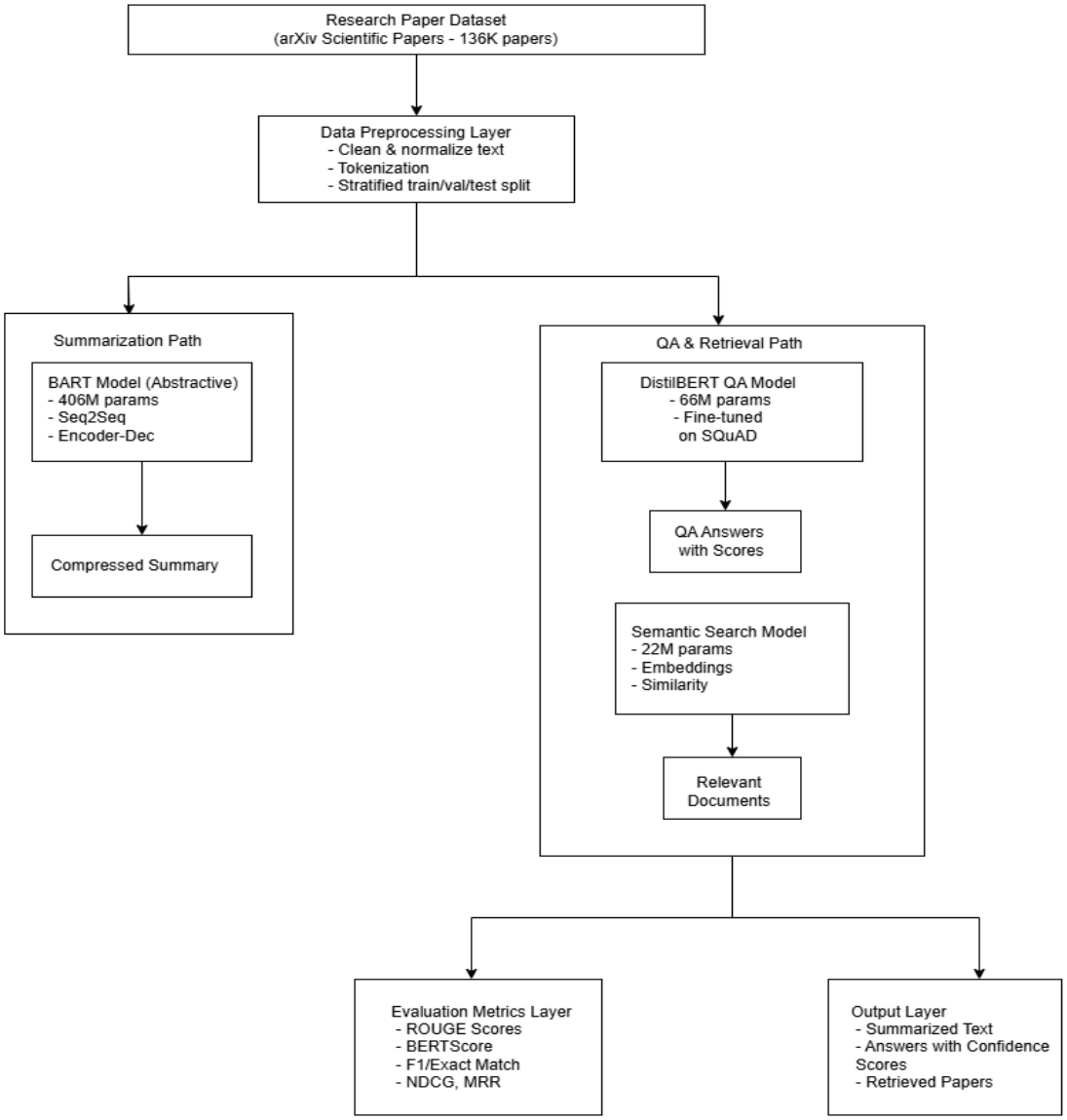
2.3.4.2 Handling Large Datasets

Research papers tend to be long and densely structured, which challenges standard transformer models that have input-length limitations. Techniques like chunking or long-document transformers (e.g., Longformer) may be required. Users have reported that for long scientific/biomedical text, both BART and PEGASUS require careful setup and may still truncate or lose context (Hugging Face discussion).

2.3.4.3 Preserving Technical Details

Scientific texts include domain-specific vocabulary, numerical tables, formulas, and highly nested argumentation structures. General purpose NLP models may misinterpret or omit these, leading to summaries that lack precision or miss key findings altogether. Ensuring that technical terms, numeric results and methodological nuance are correctly captured remains a significant challenge.

2.3.5 Proposed Architecture



The above diagram illustrates the overall system architecture of the AI-Based Research Summary and Question Answering System. It depicts how various components interact to process user queries and generate summarized research outputs.

User Interface (Frontend Interaction): Users (students, researchers, lecturers) interact with the system by entering topics, asking questions, or requesting summaries of research papers. The system processes these inputs on the backend.

Data Preprocessing Layer: Once the raw data is collected, it undergoes preprocessing. This layer cleans and normalizes the text, tokenizes it, and splits the dataset into training, validation, and test sets for the NLP models. This ensures the data is in an appropriate format for the next steps.

NLP Processing Layer (Core Functionality): This is the heart of the system where advanced natural language processing (NLP) models are applied:

- Summarization Module: The BART-Base model is used for abstractive summarization, which generates compressed summaries of research papers.
- QA Module: The DistilBERT model handles question-answering tasks. It extracts answers from the summarized content based on the user's query.
- Semantic Search Module: Powered by the paraphrase-MiniLM-L6-v2 model, this module enables the system to find relevant research papers based on the semantic meaning of the user’s queries, rather than just keyword matching.

QA & Retrieval Path: The QA & Retrieval path starts with DistilBERT for question-answering, followed by the Semantic Search Model for retrieving relevant documents based on user queries.

Evaluation Metrics Layer: Once the results are generated, the system evaluates the performance using various metrics, including:

- ROUGE Scores for summarization quality
- BERTScore for semantic similarity
- F1/Exact Match for QA accuracy
- NDCG, MRR for information retrieval quality.

Backend Orchestration Layer: This layer manages user sessions, integrates NLP-generated outputs (summarized papers, answers), and handles the business logic. It also stores the processed results in the Processed Data, Summaries & Answers Database, ensuring proper management of all generated content.

Output Presentation: The final processed data, including summarized papers, direct answers to user queries, and relevant research papers, is presented to the users. The system may provide downloadable reports or exports for academic use, ensuring easy access and usability of the research outputs.

Databases:

- Raw Papers & Metadata Database: Stores unprocessed research data collected from external sources.
- Processed Data, Summaries & Answers Database: Stores the processed and finalized summaries, answers, and related analytical results.

2.4 Existing Work

Automated summarization and question answering have been widely explored in NLP to improve how humans interact with large textual datasets. Recent advances in deep learning, particularly transformer based architectures, have significantly improved the quality of both summarization and QA models. However, when applied to scientific or technical literature, these models still face challenges in maintaining factual accuracy, handling long and complex documents, and integrating summarization with question nswering in a single workflow.
This section reviews notable studies that address various aspects of automation in literature review, extractive versus abstractive summarization, QA for technical content, and integrated summarization QA frameworks.

2.4.1 Automated Literature Review and Summarization

Automation of literature reviews using NLP and transformer models has gained substantial attention in recent years. The core objective is to reduce the manual effort required in data extraction, summarization, and synthesis from large sets of academic papers.

Sundaram & Berleant (2022) conducted a comprehensive review of NLP based automation techniques for Systematic Literature Reviews (SLRs). Their study highlighted that while text mining and rule based NLP methods had been used to automate parts of the review pipeline such as screening and keyword extraction full automation remained limited, particularly in the summarization and synthesis stages. The authors emphasized the need for robust abstractive summarization systems that can generate coherent overviews from multiple research sources, rather than relying solely on extractive approaches.

N. Fateh Ali et al. (2024) extended this line of research by comparing transformer-based summarization models with Retrieval Augmented Generation (RAG) architectures. Their evaluation revealed that while transformer models like BART and PEGASUS perform well for concise summaries, RAG-based approaches excel in contextual understanding by retrieving relevant passages before generating text. However, they identified a performance cost trade-off, as RAG systems are computationally intensive and require large memory resources, which can be prohibitive for large-scale literature automation.

H. Jin et al. (2024) provided a survey on both extractive and abstractive summarization techniques applied to technical and academic texts. Their findings revealed that most existing methods rely on general-domain datasets such as CNN/DailyMail or XSum, leading to limited domain adaptation when applied to scientific corpora. They also pointed out that evaluation metrics like ROUGE may not fully capture factual consistency and semantic adequacy indicating a need for better evaluation frameworks for scientific summarization.

In summary, recent studies show substantial progress in automating literature review tasks, but comprehensive, domain-specific abstractive summarization systems capable of maintaining accuracy and contextual relevance across scientific papers remain underdeveloped.

2.4.2 Extractive vs. Abstractive Summarization Models

Existing summarization models differ fundamentally in how they condense textual information. Extractive models select key sentences or phrases directly from the source text, ensuring factual accuracy but often producing disjointed or redundant summaries. Abstractive

models, by contrast, generate novel sentences that capture the essence of the content more naturally, closely resembling human-written summaries.

A study published in AIMS Mathematics (2024) proposed an optimized extractive summarizer (ETS-NLPODL) combining Attention Convolutional Neural Networks (ACNN), Gated Recurrent Units (GRU), and metaheuristic optimization. This hybrid approach improved sentence selection accuracy and reduced redundancy compared with classical TF-IDF or LexRank methods. However, the model lacked the semantic abstraction necessary for summarizing highly technical content making it unsuitable for contexts requiring paraphrasing or synthesis across sections.

H. Jin et al. (2024) further emphasized that abstractive approaches are critical for meaningful scientific summarization because they can rephrase complex findings, merge ideas from multiple paragraphs, and maintain logical coherence. Nonetheless, they noted that abstractive models tend to hallucinate facts, particularly when summarizing unseen scientific terminology or when trained on general-domain corpora. This challenge underscores the importance of domain-adapted fine-tuning and factual verification mechanisms. Thus, while extractive techniques ensure precision, abstractive models offer interpretability and fluency both of which are crucial for summarizing research papers intended for academic audiences.

2.4.3 Question Answering Systems for Technical Documents

Question-answering systems are designed to allow users to pose natural-language queries and receive precise answers from large textual datasets. While QA has seen major advancements in open-domain contexts (e.g., SQuAD, Natural Questions), its application to technical or scientific documents remains relatively limited.

Femmer et al. (2023) developed QAssist, a question answering system tailored for requirements analysis in software engineering. The system successfully combined information retrieval and transformer-based answer extraction, providing developers with quick, accurate responses to domain-specific questions. However, QAssist did not incorporate summarization capabilities, meaning that users still had to review long contextual texts to fully understand broader findings.

In contrast, N. Fateh Ali et al. (2024) experimented with RAG + GPT-3.5 architectures for technical document QA. The retrieval component ensured contextual grounding by extracting

relevant text chunks before generation, significantly improving answer relevance and accuracy. Despite these advantages, their study highlighted high computational cost and limited domain adaptation, as fine-tuning large language models for specific academic fields (e.g., biomedical, engineering) remains data-intensive and resource-heavy.

These findings suggest that while QA technologies are advancing, integration with summarization and domain-specific tuning are essential for effective scientific knowledge extraction.

2.4.4 Integration of Summarization and QA

Few existing studies have combined summarization and QA into a unified workflow that enables both document condensation and targeted information retrieval. Such integration is essential for efficiently navigating research literature, where users may need both a high-level overview and answers to specific technical questions.

Sundaram & Berleant (2022) emphasized the lack of end-to-end automation pipelines in Systematic Literature Review processes, noting that while some tools extract data automatically, few systems can both summarize findings and support question-driven exploration. Similarly, Femmer et al. (2023) highlighted that existing QA tools could greatly benefit from integrated summarization components to enhance research comprehension and productivity.

Combining these insights, it becomes clear that integrated summarization QA frameworks could dramatically improve how researchers interact with large volumes of academic text. Such systems would allow users to first obtain concise summaries of papers and then ask specific questions about methods, limitations, or future work all within a single interface. Despite clear demand, no widely adopted system currently provides this functionality for scientific or academic literature, leaving a notable research gap that motivates the present study.

2.4.5 Research Gaps Identified

2.4.5.1 Lack of Integrated Summarization + QA Systems for Research Literature

While summarization and question-answering (QA) have been independently explored, very few systems combine these capabilities into a single pipeline that can summarize papers and

answer user specific queries. Most existing tools focus solely on summarization (e.g., Sundaram & Berleant, 2022) or QA (e.g., Femmer et al., 2023), but not both. This separation limits the usefulness of such tools in academic research, where users need both concise summaries and precise answers derived from the same documents. Developing an integrated summarization + QA framework would allow users to efficiently navigate large sets of academic papers and extract relevant insights without manual review.

2.4.5.2 Need for Lightweight, Domain-Adapted Transformer Models

Current transformer models such as BART, PEGASUS, and T5 perform well on general domain text but often struggle with domain specific scientific terminology and structure. Research by Yu et al. (2021) and Laskar et al. (2022) indicates that fine-tuning on specialized datasets (e.g., biomedical or engineering corpora) significantly improves model relevance and factuality. However, most state-of-the-art models are computationally heavy, making them impractical for researchers with limited hardware resources. There is therefore a need for lightweight, domain-adapted transformer models for instance, fine-tuned or distilled versions of BART that maintain high accuracy while reducing computational cost.

2.4.5.3 Reproducibility Issues and Dataset Limitations

Many published studies fail to provide clear details about preprocessing, dataset composition, and fine-tuning protocols, making it difficult for others to replicate their work.

Laskar et al. (2022) and Fabbri et al. (2021) both note that limited dataset documentation and inconsistent annotation standards reduce the reproducibility of summarization experiments. Additionally, publicly available datasets such as arXiv or SciTLDR cover only a subset of scientific domains, leaving gaps in disciplines such as engineering and social sciences. This highlights the need for well-documented, domain-diverse datasets and reproducible benchmarking practices.

2.4.5.4 Evaluation Gaps: Over-Reliance on ROUGE Metrics

Most existing research evaluates summarization quality using ROUGE or BLEU scores, which measure word overlap but fail to capture factual accuracy or coherence.

As Aharoni et al. (2023) and Fabbri et al. (2021) emphasize, a summary may achieve a high ROUGE score while still containing factual inconsistencies (“hallucinations”).

For research literature, where factual precision is critical, evaluation frameworks should include QA-based metrics or human evaluation of factual correctness. Hence, there is a clear

need for multi-dimensional evaluation frameworks that assess not only lexical similarity but also factuality, coherence, and domain relevance.

2.5 Dataset Selection

2.5.1 Dataset Overview

The dataset selected for this project is the arXiv research paper dataset, a large-scale, open access collection of scientific literature widely used in the field of Natural Language Processing (NLP) and Machine Learning research.

arXiv is maintained by Cornell University and hosts millions of research papers across multiple disciplines, including computer science, mathematics, physics, engineering, and artificial intelligence. The repository's structured metadata and availability of full-text documents make it a suitable and reliable source for developing and evaluating summarization and question-answering (QA) systems.

The dataset used in this study includes metadata (e.g., titles, authors, categories, publication year) and full-text content of the papers, typically stored in PDF or JSON formats. For the purpose of this project, both the abstract and the main body text sections are extracted and processed to generate meaningful summaries and answer user-driven research questions.

2.5.2 Description of the Dataset

The arXiv dataset has become a foundational resource for academic NLP research because of its well-structured and domain rich textual data. Each entry typically contains:

1. **Title and Abstract:** Concise summaries of the research problem and contributions.
2. **Authors and Metadata:** Details that support categorization and traceability of scientific sources.
3. **Main Body Text:** Detailed sections such as Introduction, Methodology, Results, and Discussion, which provide the primary input for summarization.
4. **Categories and Subfields:** Subject-specific labels (e.g., cs.AI, cs.CL, cs.LG) that allow domain filtering and focused model evaluation.

Previous benchmark datasets such as arXiv Summarization (Cohan et al., 2018) and SciTLDR (Cachola et al., 2020) have been derived from arXiv papers. These benchmarks

have been extensively used to train and evaluate abstractive summarization models like BART, PEGASUS, and T5. Their success establishes arXiv as a standard reference corpus for summarization and scientific text analysis tasks.

2.5.3 Reason for Dataset Selection

1. **Relevance to Research Objective:** arXiv provides authentic and high quality research content, aligning directly with the project's goal of summarizing and understanding academic literature. The papers reflect current trends in technology and science, ensuring that the dataset supports both general and domain pecific applications.
2. **Scale and Diversity:** With over two million research papers across varied domains, arXiv offers a large and diverse textual corpus. This diversity allows the model to learn from a wide range of linguistic styles and terminologies, enhancing generalization and robustness. It also supports the project's plan to focus on AI and computer science papers, ensuring domain relevance.
3. **Accessibility and Open Licensing:** arXiv is freely accessible under open access policies, removing the need for commercial licenses and enabling reproducible research. Its structured API and public datasets make it easy to collect and preprocess data programmatically, supporting ethical and transparent data usage.
4. **Standard Benchmarking Potential:** Since many existing summarization datasets (e.g., SciTLDR, arXiv Summarization Corpus) are derived from arXiv, it provides a consistent benchmark for comparing results. Using arXiv ensures compatibility with prior studies, enabling direct performance evaluation against established baselines.

2.5.4 Data Preprocessing

To prepare the dataset for summarization and QA tasks, several preprocessing steps are applied to ensure data quality, consistency, and model readiness:

1. **Text Extraction:** Full-text articles are extracted from arXiv's PDF or structured JSON formats using automated text parsing tools such as pdfminer.six or the official arXiv API. The abstracts and body sections are retained for further processing.
2. **Data Cleaning:** The extracted text is cleaned to remove unnecessary elements such as:
    - References and citations (e.g., "[1]", "[2]")

- Mathematical equations, LaTeX syntax, and tables
- Figures, captions, and non-textual content
- Formatting artifacts (e.g., line breaks, special characters)

This cleaning process ensures that the model receives clear, coherent sentences suitable for summarization and QA.

3. **Sentence Segmentation and Tokenization:** Cleaned text is segmented into sentences and tokenized using spaCy or NLTK libraries to create manageable input chunks. This is particularly important for transformer-based models like BART and T5, which have a fixed maximum token limit per input sequence.
4. **Domain Filtering:** To maintain focus and relevance, papers are filtered by subject category specifically Computer Science (cs.AI, cs.CL) and Artificial Intelligence (AI) related domains. This ensures that the dataset is aligned with the project's goal of assisting students and researchers in AI-based academic areas.
5. **Final Formatting:** The cleaned and tokenized text is then formatted into structured JSON or CSV format containing fields such as:
    - Paper_id
    - Title
    - Abstract
    - Body text
    - summary (optional)
    - domain_category

This preprocessed dataset serves as the foundation for both abstractive summarization and question answering model development. It ensures that input text is high quality, domain relevant, and efficiently structured for model training and testing.

2.6 Benchmarking and Evaluation

Benchmarking and evaluation form a crucial part of assessing how effective automatic summarization and question answering (QA) systems are. In the context of this project, we compare a variety of models including traditional extractive summarizers, transformer based abstractive models, and hybrid retrieval augmented generation (RAG) systems. The evaluation framework thus spans metrics for summarization, QA performance, computational efficiency, and real world usability.

2.6.1 Evaluation Metrics for Summarization & QA Models

2.6.1.1 Accuracy, Precision, and Recall

When evaluating summarization or QA systems, the metrics of accuracy, precision, and recall are foundational.

- **Accuracy** reflects how often the system's output (summary or answer) is correct relative to a reference.
- **Precision** indicates the proportion of relevant content in the system's output.
  e.g., in a summary, how much of what was generated is actually relevant to the source text or answer context.
- **Recall** measures how much of the key or expected information from the source was captured in the summary or answer.
  In QA systems, high precision and recall show that the system retrieves and correctly answers user queries with relevant context. Some recent comparative studies find that transformer based models (for example, a fine-tuned T5-base or GPT-3.5-turbo with RAG) yield higher precision and recall than older extractive models such as those built on ACNN-GRU optimized extractive architectures (AIMS Mathematics, 2024) and other hybrid systems.

2.6.1.2 F1-Score and AUC-ROC

The **F1-score** integrates precision and recall into a single metric (their harmonic mean), providing a balanced measure especially useful when precision and recall are both important. The **AUC-ROC** (Area Under the Receiver Operating Characteristic curve) shows how well a model distinguishes relevant from irrelevant content (for example, correct vs incorrect answers or coherent vs incoherent summary segments). Studies of hybrid systems report that abstractive transformer models such as T5 and GPT variants achieve higher F1 and AUC scores compared to purely extractive or simpler summarization methods (Ali et al., 2024; Jin et al., 2024).

2.6.1.3 Computational Efficiency and Processing Time

Beyond correctness, in practical use cases efficiency matters. Extractive summarization models (e.g., ETS-NLPODL) tend to require less compute, process faster, and may work on more modest hardware, but often produce summaries that are less fluent, less coherent or

semantically rich. Meanwhile, abstractive transformer-based models (like T5, PEGASUS, or GPT-based systems) demand more computational resources (GPU memory, longer training/inference time) but produce richer, human-like summaries. The trade-off between speed, resource use, and output quality is routinely highlighted (Sundaram & Berleant, 2022). For large-scale literature review automation (processing many full research papers), both efficiency and interpretability become critical.

2.6.2 Comparative Analysis of Existing Models

2.6.2.1 Benchmarking Studies

In benchmarking summarization systems, extractive models (like ETS-NLPODL) often perform strongly on n-gram overlap metrics (e.g., ROUGE) but lack deeper semantic abstraction and coherence across longer texts. Abstractive transformer models (T5, BART, PEGASUS) provide better coherence and paraphrasing ability, but at higher cost and sometimes with increased risk of factual errors or "hallucinations". Hybrid RAG (Retrieval-Augmented Generation) systems which retrieve relevant passages and then generate summaries or answers aim to combine context awareness with generation fluency, thereby improving factual grounding and supporting QA integration albeit requiring extra system complexity and computational resources.

2.6.2.2 Real-World Performance and User Experience

In applied settings, QA systems such as QAssist (Femmer et al., 2023) show that retrieval-based QA works well for structured documents, but adapting them to full research literature is more challenging due to domain-specific language, long texts, and subtle methodological details. Users often prefer tools that produce context-aware abstractive summaries combined with interactive QA, enabling them to not only get a short summary but also ask specific questions (e.g., about limitations, future work) within the same workflow. This user preference reinforces the rationale of the present project's approach, which integrates summarization + QA in one platform.

2.6.3 Challenges in Benchmarking Multimodal Systems

Several persistent challenges complicate benchmarking of summarization + QA systems:

- **Lack of Standard Datasets:** There is currently no widely accepted unified benchmark that covers full-length research-paper summarization + QA in academic

domains. Many studies focus on news or general domain corpora, making cross model comparison difficult for the research-paper domain.

- **Subjectivity of Evaluation:** Human evaluation of summaries (e.g., readability, coherence, relevance) is expensive, time-consuming and subjective. Reliance on automatic metrics such as ROUGE alone does not guarantee quality. Recent work shows ROUGE has low correlation with factual correctness or user satisfaction.
- **Balancing Speed and Depth:** Higher performance models often incur higher compute cost or longer processing time, which can be problematic for large scale use or limited hardware contexts.
- **Domain Adaptability:** Models trained on general text (news, web) often under-perform on scientific literature, which uses specialized vocabulary, multi-section structure and dense arguments. This mismatch limits the effectiveness of many state-of-the-art systems when applied in academic domains.
- **Beyond Overlap Metrics – Factuality & QA-based Evaluation:** A key limitation is that many summarization evaluations rely on overlap-based metrics such as ROUGE (n-gram overlap) which do not account for factual consistency or whether the summary or answer is correctly grounded in the source document. Research into factual consistency metrics (e.g., QAFactEval, QAGS) shows this gap. For example, QAGS uses question generation and answering on summary vs source to detect factual inconsistencies. Hence, future benchmarking must consider both quantitative metrics (e.g., ROUGE, BLEU, F1, AUC) and qualitative/grounded metrics (factual accuracy, coherence, user satisfaction, QA-based measures) to enable a comprehensive evaluation of system performance.

2.7 Chapter Summary

This chapter reviewed existing literature on automated summarization and QA systems for research papers, revealing significant advancements from traditional extractive models. The review identified key research gaps, including the lack of integrated summarization and QA systems tailored for scientific literature, insufficient domain adaptation, the absence of standardized datasets and balanced evaluation metrics, and the need for lightweight yet effective transformer models suitable for real world deployment. These insights collectively justify and support the proposed research, which seeks to develop an integrated, domain-adapted summarization and QA model designed to deliver accurate, context-aware, and computationally efficient understanding of research literature.

Chapter 03: Methodology

3.1 Chapter Overview

This chapter outlines the methodology adopted for developing this project. It explains the research and design approaches used to achieve the project objectives, including data collection, preprocessing, model training, testing, and evaluation. The chapter also discusses

the chosen system design methodologies, programming paradigms, and testing strategies that guide the development process. Additionally, it provides detailed insights into the project management framework, covering scope, schedule, resources, and risk assessment. Through these structured methodologies, the chapter ensures a clear and systematic approach to building an efficient and reliable AI-driven research assistant.

3.2 Research Methodology (Saunders' Research Onion in Table Form)

Layer	**Choice**	**Justification**
Philosophy	Pragmatism	This project uses both quantitative and qualitative methods. Quantitative analysis is used for evaluating AI model performance, while qualitative insights are gathered through user surveys to understand user satisfaction and ease of use. Pragmatism supports combining both methods to find the most practical solution to the research problem.
Approach	Deductive Approach	A deductive approach is used because the project tests a specific hypothesis that an AI-based research assistant can accurately summarize papers and answer questions. This involves building a model based on existing theories and then testing it with real data to confirm its effectiveness.
Methodological Choice	Mixed methods (Quantitative and qualitative)	The study includes both quantitative (AI model evaluation, accuracy scores) and qualitative (user feedback through surveys) data. This combination helps provide a more complete understanding of the system's performance and user experience.

Strategy	Experimental and Survey	**Experimental:** Used to test and evaluate the AI model's performance on summarization and question answering tasks using research paper datasets. **Survey:** Conducted with users to collect feedback about system usability and usefulness. Since the survey is structured with closed-ended questions, it follows a quantitative approach.
Time Horizon	Cross-sectional	The research is conducted over a short period of time using data collected at one specific stage of development (a single snapshot). This means data are not collected repeatedly over time, which makes the study cross-sectional.
Data Collection	Secondary Data	Existing public research papers and datasets were used to train and test the AI model. These datasets were collected from trusted online sources like Google Scholar and Semantic Scholar. Since the data already exists and is not collected directly from participants, it is secondary data.

3.3 Development Methodology

Software development methodologies define the process used to plan, design, build, test, and deploy a system. Choosing the right development methodology is important to ensure that the system is delivered efficiently and meets user requirements.

Chosen Development Methodology: Agile Methodology

Justification

The Agile methodology was chosen for this project because it allows flexibility, adaptability, and continuous improvement during development. The project involves developing an

AI-based research assistant that integrates web scraping, summarization, and question-answering modules each of which can be developed and tested independently in stages.

Agile supports incremental development, where each component can be improved based on testing results and user feedback. For example, after the first sprint, the summarization feature was tested and refined before integrating the question-answering module. Agile also encourages frequent communication and testing, which ensures that the final system meets user expectations in terms of performance and usability.

In summary, Agile was selected because it:

- Allows flexible changes during development.
- Supports frequent testing and integration.
- Improves collaboration and feedback handling.
- Is suitable for AI/ML-based systems that require experimentation and tuning.

3.3.1 Requirement Elicitation Methodology

Chosen Requirement Elicitation Method: Online Survey

Justification

For this project, an online survey was used to collect feedback from potential users primarily students and researchers. The survey focused on identifying what users expect from an AI-based research tool, including desired features, ease of use, and the usefulness of automatic summarization and question-answering capabilities.

Since the survey contained closed-ended questions, it produced quantitative data, making it easier to analyze and interpret statistically. This method was selected because:

- It allows collecting responses from a large number of users efficiently.
- It provides measurable results that can be used to prioritize features.
- It helps understand user preferences without the need for face-to-face meetings.
- It is cost-effective and suitable for online academic research environments.

The data gathered from the survey helped define both functional requirements (e.g., ability to upload research papers and generate summaries) and non-functional requirements (e.g., system performance, usability, and response speed).

3.3.2 Design Methodology

System design methodology defines how the structure and functionality of the system are analyzed and represented before development begins. Common design methodologies include Structured Systems Analysis and Design Methodology (SSADM) and Object Oriented Analysis and Design Methodology (OOADM).

1. Structured Systems Analysis and Design Methodology (SSADM)

SSADM is a traditional design approach that focuses on processes and data flow. It breaks the system into smaller functional parts using tools such as Data Flow Diagrams (DFDs), Entity Relationship Diagrams (ERDs), and Logical Data Models. This methodology is useful for systems where the main focus is on data movement and processing rather than modular structure.

2. Object-Oriented Analysis and Design Methodology (OOADM)

OOADM focuses on identifying objects, their attributes, behaviours (methods), and interactions within the system. It uses UML (Unified Modeling Language) diagrams such as use case diagrams, class diagrams, sequence diagrams, and activity diagrams to visualize system functionality. OOADM is well-suited for modern software systems built using object-oriented programming languages such as Python or Java.

Chosen Design Methodology: Object Oriented Analysis and Design Methodology (OOADM)

Justification

The Object-Oriented Analysis and Design Methodology (OOADM) was selected for this project because the system is developed using Python, which supports object-oriented programming principles such as encapsulation, inheritance, and modularity.

This approach allows the system to be divided into independent modules such as:

- **Web Scraper Module:** Handles data collection from research paper sources.
- **Summarization Module:** Processes text data and generates concise summaries using NLP models.
- **Question Answering Module:** Interprets user queries and extracts relevant answers.
- **User Interface Module:** Provides an interactive interface for users to input topics, view summaries, and ask questions.

By applying OOADM:

- Each module can be designed, tested, and maintained separately.
- The system becomes scalable and reusable, allowing easy integration of new AI models or data sources in the future.
- UML diagrams such as use case diagrams and class diagrams clearly represent system functionality and data interaction.

Hence, OOADM provides a clear and systematic approach for designing the AI-based application, ensuring both technical efficiency and logical clarity in system development.

3.3.3 Programming Paradigm

A programming paradigm is a style or way of organizing and structuring code to solve problems effectively. Choosing the right programming paradigm helps ensure that the system is efficient, easy to maintain, and scalable. The most commonly used paradigms include Object Oriented Programming (OOP), Structured Programming, and Functional Programming.

1. Object-Oriented Programming (OOP)

OOP is based on the concept of objects, which combine data (attributes) and behaviour (methods) into single reusable components. It supports key principles such as encapsulation, inheritance, abstraction, and polymorphism. This approach promotes modular design, code reusability, and easy maintenance, making it ideal for large or complex systems. OOP is commonly implemented using languages such as Python and Java.

2. Structured Programming

Structured programming follows a top-down approach, focusing on logical control flow using loops, conditions, and functions. It is suitable for procedural tasks that require clear step-by-step processing. This paradigm improves readability and debugging but may become complex for large scale systems that need modular and reusable components.

3. Functional Programming

Functional programming treats computation as the evaluation of mathematical functions and avoids changing states or mutable data. It is highly useful for tasks involving complex data transformations and parallel processing. Languages such as Haskell, Scala, and R are commonly used for this paradigm.

Comparison of Paradigms

Structured programming emphasizes control flow, while functional programming focuses on immutability and mathematical transformations. In contrast, object-oriented programming allows the system to be divided into smaller, reusable, and self-contained modules that can interact with one another. This makes OOP more suitable for modular and extensible applications like AI-based systems.

Chosen Programming Paradigm: Object-Oriented Programming (OOP)

Justification

This project adopts the Object-Oriented Programming paradigm because it aligns perfectly with the modular design of the system. The application consists of several independent yet connected modules such as:

- **Web Scraping Module –** collects relevant research papers from online sources.
- **Summarization Module –** processes and summarizes text using NLP techniques.
- **Question Answering Module –** interprets user queries and retrieves relevant answers.
- **User Interface Module –** provides interaction between the system and the user.

Using OOP principles allows these modules to be developed, tested, and maintained independently, while still working together as a unified system. It also improves code reusability, scalability, and readability, which are important for AI-driven applications developed in Python.

3.3.4 Testing Methodology

Testing is an essential phase in software development used to verify that the system functions correctly, meets user requirements, and performs efficiently under different conditions. In this project, two main testing approaches were adopted Model Testing and Prototype Testing to ensure both the individual components and the complete system perform as expected.

3.3.4.1 Model Testing

Model testing focuses on evaluating the accuracy and reliability of the machine learning or NLP models used in the system. Each model, such as the text summarization model and the question answering model, was tested using standard performance metrics, including:

- **Accuracy –** Measures how often the model produces the correct output.
- **Precision –** Evaluates the proportion of correctly predicted positive results.
- **Recall –** Determines how well the model identifies all relevant results.
- **F1-Score –** Combines precision and recall into a single balanced measure.

This testing ensured that the underlying models were capable of generating relevant, coherent, and contextually accurate summaries and answers before being integrated into the main system.

3.3.4.2 Prototype Testing

Prototype testing involves evaluating the overall functionality, integration, and usability of the complete system. Once all modules were connected including text extraction, summarization, question-answering, and the user interface the prototype was tested to ensure smooth interaction between components and positive user experience.

The focus areas included:

- Verifying that each module communicates correctly with others.
- Checking the response time and system stability.
- Gathering feedback from users to identify areas for improvement.

This phase ensured that the system performed effectively not only from a technical perspective but also in terms of user satisfaction and practical usability.

By combining Model Testing for individual model performance and Prototype Testing for overall system evaluation, the project achieved a balanced and thorough testing process. This dual approach helped ensure that the AI-based research assistant system is both technically accurate and user-friendly, meeting the objectives defined in the project requirements.

3.3.5 Solution Methodology

The solution methodology defines the step-by-step process followed to develop this project. The approach combines NLP and machine learning techniques to collect, summarize, and answer questions from research papers. The main stages include Dataset Collection, Data Preprocessing, Feature Engineering, Model Selection and Training, Testing, and a Feedback Loop for continuous improvement.

3.3.5.1 Dataset Collection

The dataset for this project was created by collecting research papers and scientific documents from online sources such as Google Scholar, Semantic Scholar, and research repositories. Each paper was extracted in text form, including its title, abstract, introduction, methodology, results, and conclusion. This dataset provided the input text needed for both summarization and question answering tasks.

3.3.5.2 Data Preprocessing

Before training or using the model, the collected text data was cleaned and preprocessed to ensure consistency and readability. The preprocessing steps included:

- Removing unnecessary elements such as citations, references, and special characters.
- Converting all text to lowercase for uniformity.
- Tokenizing sentences and words for better model understanding.
- Removing duplicate or irrelevant text sections.

This stage ensured that only meaningful and clean textual data was used for analysis and model input.

3.3.5.3 Feature Selection and Engineering

In this project, the main “features” are text-based elements such as:

- Research paper titles, abstracts, and main content sections.
- Keywords and phrases related to important concepts.
- Contextual relationships between questions and the corresponding parts of the paper.

Feature engineering involved structuring these textual elements so that the model could effectively understand context, summarize key points, and generate accurate answers.

3.3.5.4 Model Selection

The BART-base model was chosen for this project. BART is a transformer based NLP model developed by Facebook AI that performs exceptionally well on text summarization and question answering tasks. It combines the strengths of BERT (bidirectional encoder) and GPT (auto-regressive decoder), enabling it to understand the context of text and generate high-quality summaries and answers.

This model was selected because it:

- Handles long and complex academic text efficiently.
- Generates coherent and human-like summaries.
- Provides context-aware and concise answers to user queries.

3.3.5.5 Model Training

The BART-base model was fine-tuned using a research paper dataset that included paper abstracts, sections, and example question answer pairs.

During training:

- The model learned to generate short, informative summaries for each paper.
- It also learned to answer specific questions (e.g., “What are the limitations?” or “What is the future work?”) by understanding context from the input text.

The training process involved optimizing model parameters to minimize loss and improve accuracy in both summarization and question answering tasks.

3.3.5.6 Testing

After training, the model was tested using unseen research papers to evaluate its performance. Performance metrics used included:

- **ROUGE Score –** To measure the similarity between generated summaries and reference summaries.
- **BLEU Score –** To assess the quality and fluency of generated text.
- **Accuracy and Relevance –** For question-answering results, evaluated based on how relevant and correct the answers were.

This testing ensured that the model produced accurate, concise, and meaningful summaries and answers.

3.3.5.7 Feedback Loop

A feedback loop was implemented to improve model performance continuously.
User feedback and evaluation results were analyzed to identify:

- Cases where summaries were too long or missed important points.
- Instances where answers were incomplete or irrelevant.

Based on these findings, the model parameters and preprocessing steps were refined. This iterative improvement process ensured that the system became more reliable and user-friendly over time.

Overall, this solution methodology combines modern NLP techniques and transformer based deep learning to build a robust AI system. Using BART-base allows the project to generate accurate summaries and contextual answers, making it a powerful academic research assistant for students and researchers.

3.4 Project Management Methodology

Effective project management ensures that the system is developed, tested, and evaluated within the planned time, budget, and quality constraints.
For this project, a phased project management approach based on the Agile methodology was followed. Agile allows flexibility, iterative development, and continuous feedback which is ideal for AI-based systems that require model training, testing, and refinement.

Each stage of the project was organized into manageable tasks such as requirement gathering, design, implementation, testing, and evaluation, with regular progress reviews to ensure timely completion.

3.4.1 Scope

3.4.1.1 In-Scope

- Developing an AI-based application capable of summarizing research papers and answering user queries.

- Implementing web scraping or data collection to retrieve research papers related to a user-defined topic.
- Using the BART-base model for generating concise summaries and context-aware answers.
- Building a user interface (UI) that allows users to input topics, view summaries, and ask questions.
- Testing the system using sample research papers to evaluate its accuracy, relevance, and usability.

3.4.1.2 Out-Scope

- Deploying the system as a large scale web or mobile application for public use.
- Integrating payment gateways, user accounts, or cloud hosting features.
- Creating multilingual or domain specific versions of the model.
- Conducting real time web scraping from restricted or paid academic databases.
- Long term maintenance or commercialization of the application.

3.4.2 Schedule

3.4.2.1 Gantt Chart

3.4.2.2 Deliverables

Delivarebles	**Dates**
Project Proposal	8th Oct 2025
Literature Review	9th Oct 2025
Project Proposal	10th Oct 2025
Project Proposal and Requirement Specification – final draft	12th Nov 2025
Project Proposal and Requirement Specification – Final	13th Nov 2025
Design Document	15th Dec 2025
Prototype	5th Feb 2026
Interim Project Demo	6th Feb 2026
Implementation	7th Feb 2026
Testing	8th Feb 2026

Evaluation	15th Feb 2026
Thesis Submission	30th Mar 2026
Prototype	30th Mar 2026
Dataset and Source Code	30th Mar 2026

3.4.3 Resource Requirements

The successful development and testing of the AI-Based Research Summary and Question Answering Application require several key resources, including hardware, software, data, and technical skills. These resources ensure that the system performs efficiently and meets its intended objectives.

3.4.3.1 Hardware Requirements

The project requires a system with moderate to high computational capacity to support text preprocessing, model fine-tuning, and inference tasks.

Hardware Specifications:

- Processor: Intel Core i7 or higher (or equivalent AMD processor)
- Memory (RAM): Minimum 16 GB
- Storage: Minimum 512 GB SSD
- GPU (optional but recommended): NVIDIA GPU with CUDA support for faster model execution

These specifications are necessary for handling large research paper text files and performing NLP model operations smoothly, especially when working with transformer based models like BART-base.

3.4.3.2 Software Requirements

The project uses a combination of programming tools, frameworks, and libraries for implementation, training, and testing.

Software Stack

- Programming Language: Python
- Frameworks/Libraries:

Transformers (Hugging Face) – for using and fine-tuning the BART-base model
PyTorch – deep learning framework used to implement and train the model
NLTK / spaCy – for text preprocessing (tokenization, stop-word removal, etc.)
Flask / Streamlit (optional) – for building the user interface

- Development Environment: Jupyter Notebook / Visual Studio Code

These tools collectively support the data processing, model training, testing, and user interface development stages.

3.4.3.3 Data Requirements

The project utilizes research papers and academic documents collected from open-access repositories and academic platforms such as Google Scholar, Semantic Scholar, or arXiv.

Data Characteristics:

- Textual data including titles, abstracts, and body content of research papers.
- Cleaned and preprocessed for summarization and question answering tasks.
- Used to fine-tune and evaluate the BART-base model's performance.

All datasets used are publicly available and used strictly for educational and research purposes.

3.4.3.4 Skills Requirements

The successful completion of the project requires a combination of technical and analytical skills:

Required Skills

- **Python Programming:** For implementing and integrating all components of the system.
- **NLP:** Understanding of text summarization, tokenization, and context modeling.
- **Machine Learning & Deep Learning:** Knowledge of transformer models and model evaluation.
- **Data Preprocessing & Cleaning:** Ability to prepare unstructured textual data for model input.
- **Software Development:** Skills in designing and testing the application prototype.

These skills ensure that the system is technically sound, efficient, and user-friendly.

The resource requirements encompass both technical infrastructure and specialized expertise essential for building the AI-based research assistant. Adequate allocation of these resources ensures that the project achieves high accuracy, efficiency, and usability in summarizing and answering questions from research papers.

3.4.4 Risk and Mitigation

Risk	**Severity**	**Frequency**	**Mitigation Strategy**
Insufficient training data	4	3	Use multiple open-access sources (e.g., arXiv, Semantic Scholar) to collect diverse and reliable datasets. Apply data cleaning and preprocessing to ensure quality.
Poor model accuracy	5	3	Fine-tune the BART-base model using additional datasets. Optimize hyperparameters and validate with ROUGE and BLEU metrics.
Overfitting	4	3	Use cross validation, dropout, and early stopping to avoid overfitting.
Hardware limitations	3	2	Use cloud-based platforms such as Google Colab, AWS, or Kaggle for training and testing.
High false-positive rate	3	3	Optimize the model and preprocessing pipeline. Use model compression and caching techniques.

Stakeholder dissatisfaction	3	2	Collect feedback through surveys and testing sessions to ensure the system meets user expectations.
Delays in model training	3	3	Use smaller data samples for initial runs. Optimize training using GPUs and parallel processing.
Data privacy or copyright issues	4	2	Use only open access papers and datasets. Avoid scraping restricted or paid content.

3.5 Chapter Summary

This chapter presented a comprehensive explanation of the methodologies applied throughout the project development. It detailed the research approach, system design, programming paradigms, and testing procedures adopted to ensure the accuracy and performance of the BART-based model. Furthermore, it described the solution workflow from dataset collection and preprocessing to model training, evaluation, and refinement. The project management section outlined the scope, timeline, required resources, and identified potential risks with corresponding mitigation strategies. Overall, this methodology establishes a strong foundation for the successful implementation and evaluation of the AI-based research summarization and question answering system.

Chapter 04: Software Requirement Specification (SRS)

4.1 Chapter Overview

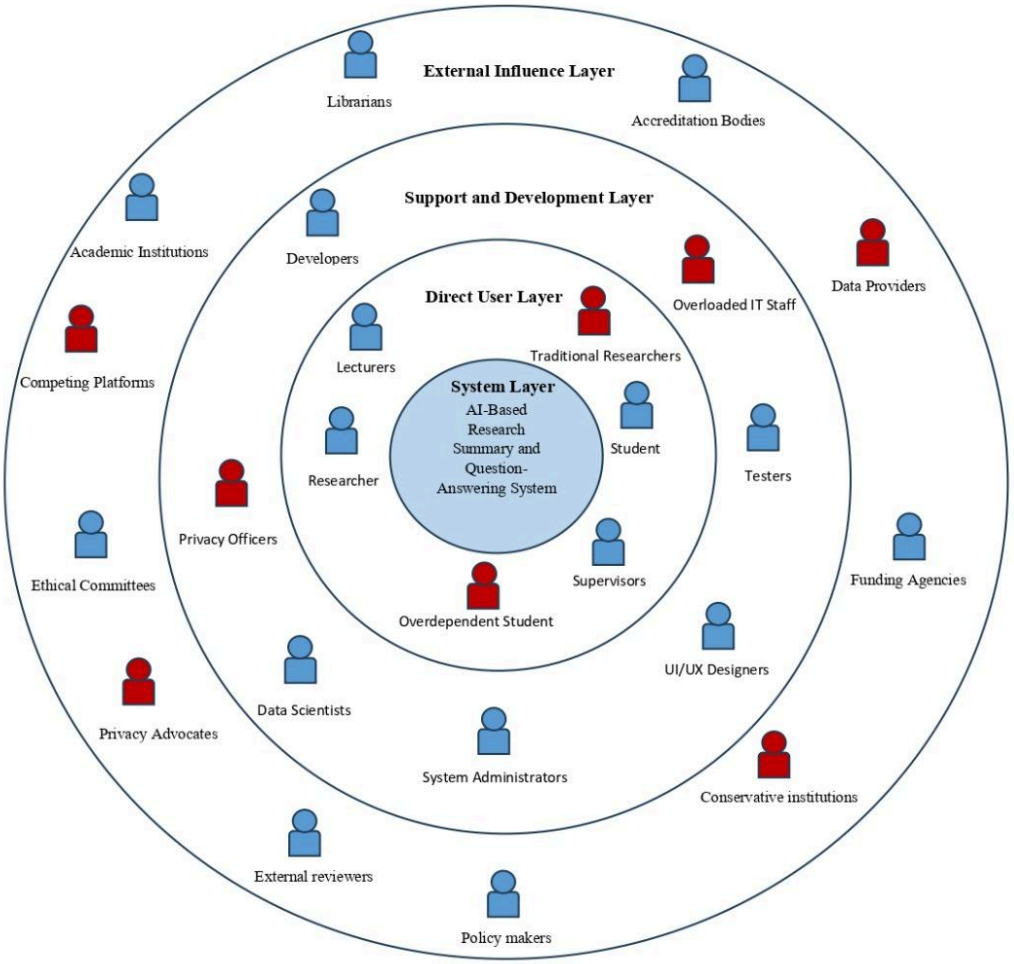
This chapter constitutes the SRS for the AI-Based Research Summary and Question Answering System. It formally defines the boundaries, scope, actors, and functional and non-functional requirements necessary for successful system development.

4.2 Rich Picture Diagram

The Rich Picture Diagram is the overview of the project and visually maps its interactions with all key actors, including Lecturers and the System Administrator. Crucially, the RPD highlights underlying complexities and risks, such as concerns over Data Reliability, API Limits from external sources, the need for Usability, and critical non-functional challenges like Scalability and mitigating Hallucination/Accuracy issues, thereby providing a comprehensive, subjective snapshot before detailed technical modeling begins.

4.3 Stakeholder Analysis

4.3.1 Stakeholder Onion Model



4.3.2 Stakeholder Viewpoints

Stakeholder	Role	Description
AI-Based Research Summary and Question Answering System	Core system	Acts as the main platform that automates paper collection, summarization, and question answering. Its accuracy and efficiency determine overall system success.
Students	Primary users	Expect fast and accurate summaries of research papers, easy navigation, and reliable answers to academic questions to support coursework and projects.
Researchers	Advanced users	Require detailed, context aware summaries and question answering features to speed up literature reviews and identify research gaps.
Lecturers / Supervisors	Academic mentors	Seek reliable summaries for verifying student work and assisting with research guidance. Expect the system to support academic integrity.
Traditional Researchers	Resistant users	May view AI summarization as oversimplifying academic work or reducing the need for in-depth reading. Concerned about the authenticity of generated summaries.
Overdependent Students	Misusing users	Might rely too heavily on the system and neglect critical thinking or manual literature analysis, affecting learning outcomes.
Developers / Data	System creators	Focused on implementing, training, and

Scientists		optimizing the model for accuracy and efficiency. Expect sufficient computing resources and data access.
System Administrators / Testers	Technical maintainers	Responsible for deployment, performance monitoring, and issue resolution. Expect stable infrastructure and security compliance.
Overloaded IT Staff	Operationally affected	May face additional workload from maintaining servers and ensuring continuous updates, possibly leading to resistance or delays.
Privacy Officers	Data compliance regulators	Concerned with data protection, licensing of research papers, and ethical handling of extracted content. Expect strict adherence to privacy laws.
Academic Institutions	Institutional adopters	Interested in integrating the system within digital learning environments to enhance research efficiency and resource accessibility.
Librarians	Information managers	Expect the system to align with library cataloging standards and improve resource accessibility for students and faculty.
Accreditation Bodies / Policy Makers	Regulators	Ensure the system aligns with academic and ethical standards, promoting fair use of AI-generated summaries.
Funding Agencies	Project supporters	Focused on the system's research impact, scalability, and contribution to academic innovation. Expect clear

		project outcomes.
Ethical Committees / External Reviewers	Oversight and quality assessors	Ensure compliance with research ethics, data transparency, and fairness in AI-generated outputs.
Competing Platforms	Rival organizations	May see the system as a threat to existing academic databases or subscription-based research tools.
Data Providers	Data source owners	Concerned about potential misuse or scraping of research content without proper licensing.
Privacy Advocates	Ethical critics	Worry about data privacy, intellectual property, and potential misuse of generated summaries.
Conservative Institutions	Adoption barriers	Hesitant to integrate AI-driven tools due to skepticism about reliability and academic authenticity.

4.4 Selection of Requirement Elicitation Methodologies

To ensure that the system requirements were accurate, user centered, and aligned with academic needs, several requirement elicitation techniques were used. These methods helped gather insights from key stakeholders such as students, researchers, and lecturers.

Methodology	**Description**	**Purpose**
Literature Review	Reviewed recent studies	To establish a technological

	(2017–2025) related to AI-based summarization, academic question-answering systems, and NLP applications in research support tools.	foundation and identify gaps in existing research summarization and Q&A systems.
Survey	Distributed a structured online Google Form to students, lecturers, and researchers. Collected both quantitative and qualitative data on system usability expectations, preferred summary formats, and trust in AI-generated research content.	To capture user needs, interface preferences, and identify common challenges in academic research summarization and information retrieval.
Document Analysis	Analyzed academic paper metadata, citation formats, and dataset documentation standards from IEEE, Springer, and Elsevier.	To determine standard structures for data extraction, citation accuracy, and summarization consistency in the system.

4.5 Discussion of Findings

4.5.1 Literature Review Findings

- Most existing research summarization and academic Q&A systems lack domain adaptability and struggle to handle complex scientific language.
- Transformer based models (such as T5, BERT, and GPT variants) have shown strong performance in text summarization and question answering, but they require high quality and domain specific datasets for accurate results.
- Studies show that integrating knowledge graphs and metadata driven retrieval techniques enhances the interpretability and contextual accuracy of AI-generated summaries (Sundaram & Berleant, 2022).

- Many current systems focus only on generic document summarization, overlooking the academic structure of research papers such as abstracts, methods, results, and conclusions.
- There is a growing demand for systems that support researchers, students, and lecturers in efficiently retrieving, summarizing, and understanding scientific literature.

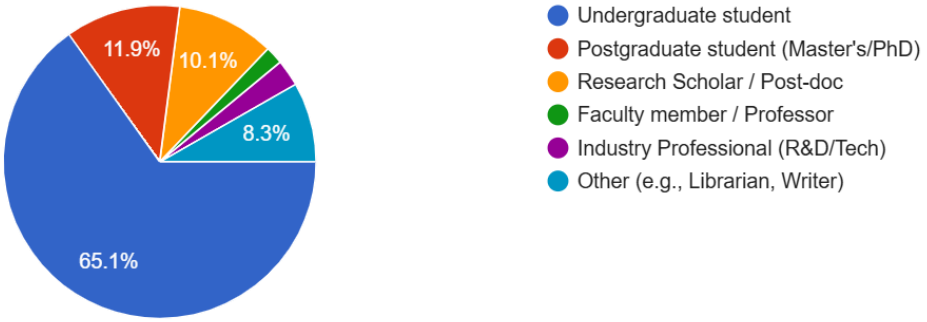
4.5.2 Interviews Findings

Codes	**Themes**	**Conclusion**
Information Overload	Difficulty in manually reading and summarizing large volumes of research papers	The system should automate summarization and highlight key findings for quicker understanding.
Answer Accuracy	Variations in response quality from existing AI tools	A reliable question answering module is needed to ensure contextually accurate and concise answers.
Usability	Preference for intuitive and organized interfaces	The system should include a simple dashboard with categorized search results and clear visualization.
Data Access	Limited access to full-text papers and metadata	Integration with external data sources (e.g., open access repositories) is necessary for comprehensive analysis.

4.5.3 Survey Findings

1.What is your current role/educational status?

109 responses

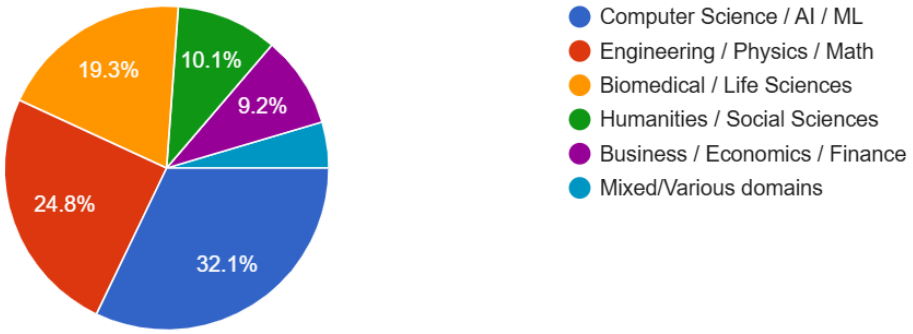


2.How experienced are you with reading technical research papers?

109 responses

3.Which domain of research papers do you primarily read?

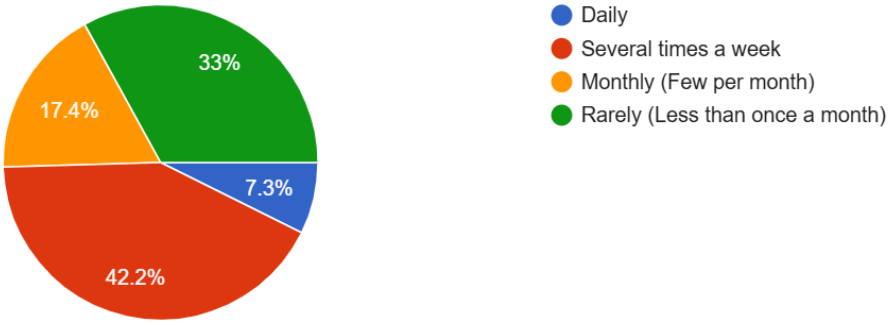
109 responses



4.How often do you typically read research papers?

Copy c

109 responses



5.Where do you most frequently search for research papers?

109 responses

Section 2: Challenges & Pain Points

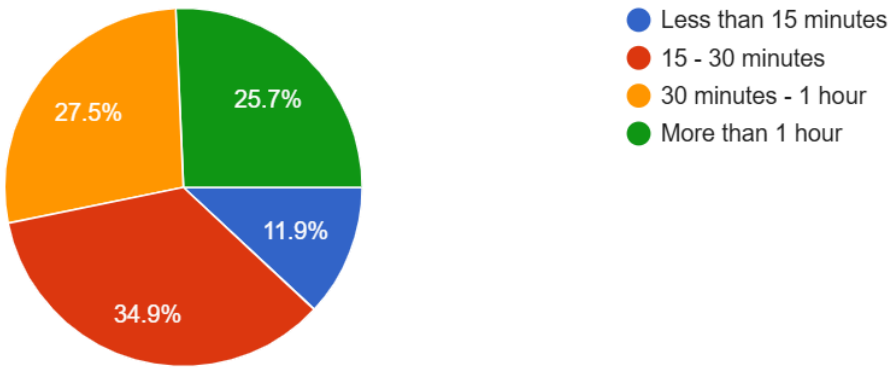
1.What is your biggest obstacle when reading or trying to understand a new paper?

Copy chart

109 responses

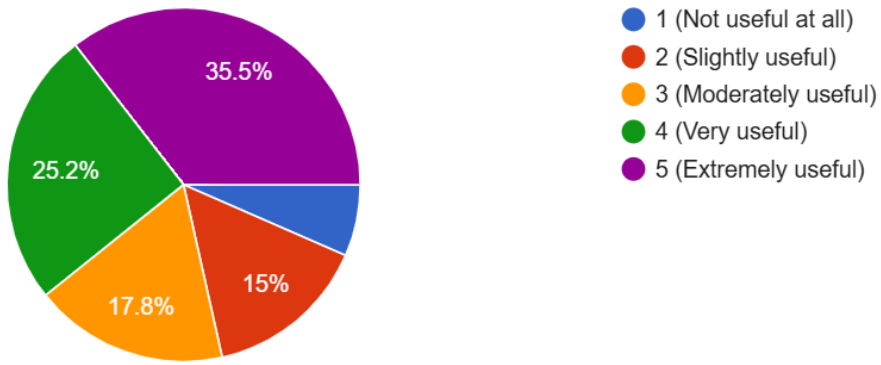
2.How much time do you usually spend just trying to understand the core arguments of a single paper?

109 responses



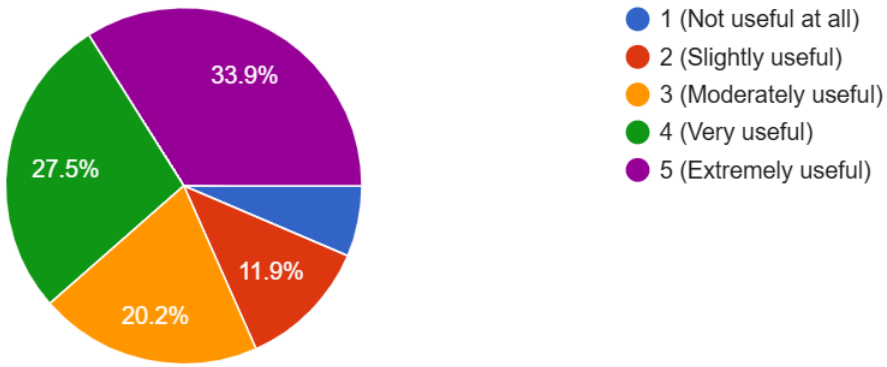
3.How useful would an AI tool be that automatically summaries research papers for you?

107 responses



4.How useful would it be if the tool could also answer specific questions about a paper (e.g., limitations, methods, or future works)?

109 responses



Section 3: Feature & Output Preferences

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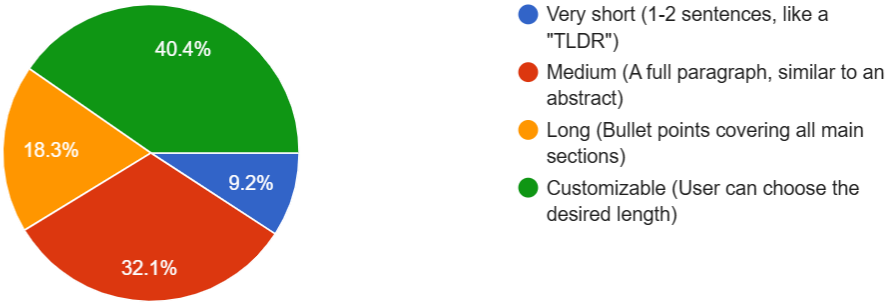
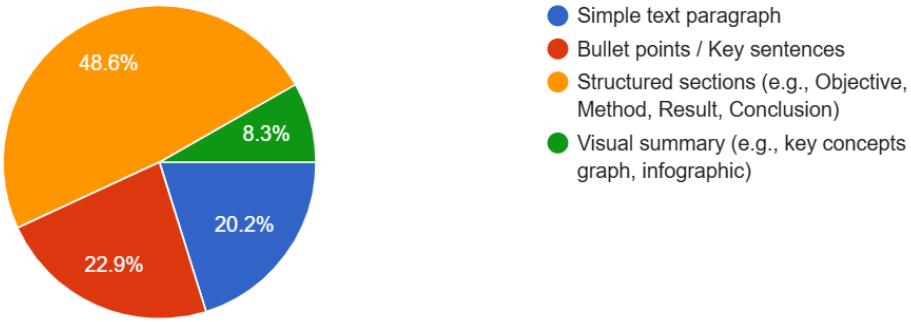
1.What summary length would be most helpful to you?

109 responses



2.Which format would you prefer for the summary output?

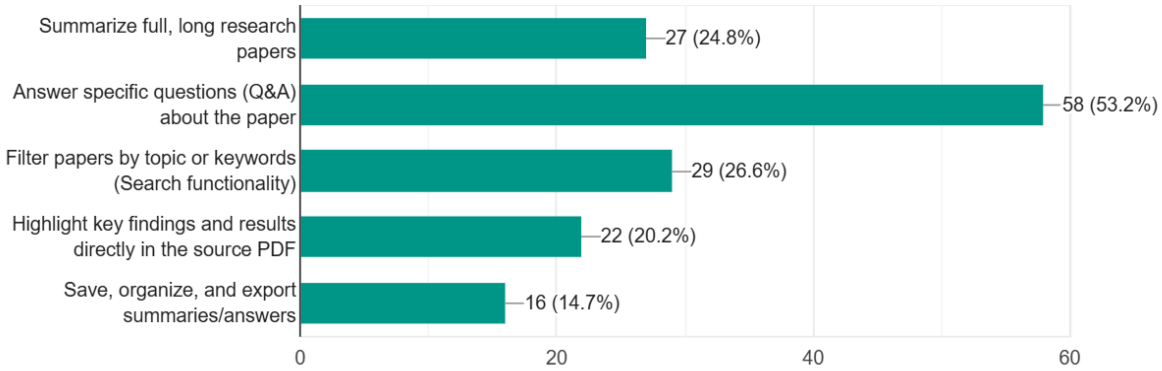
109 responses



3.Which features are most essential for you in an AI research app?

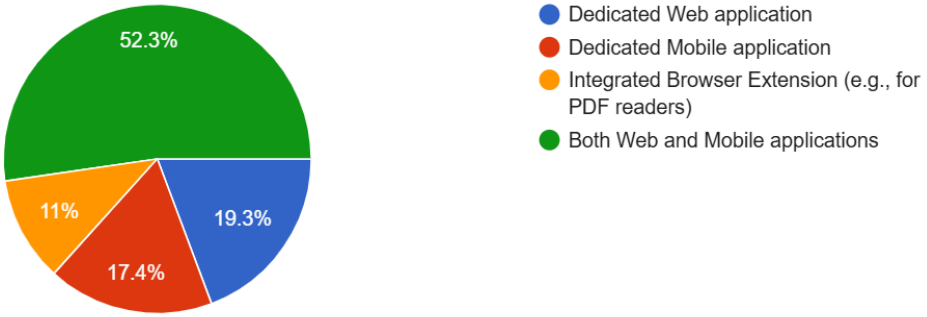
Copy chart

109 responses



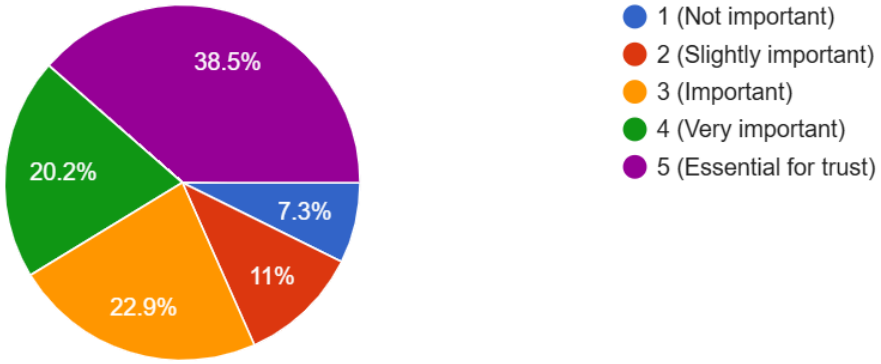
4.Would you prefer to use this tool as a:

109 responses



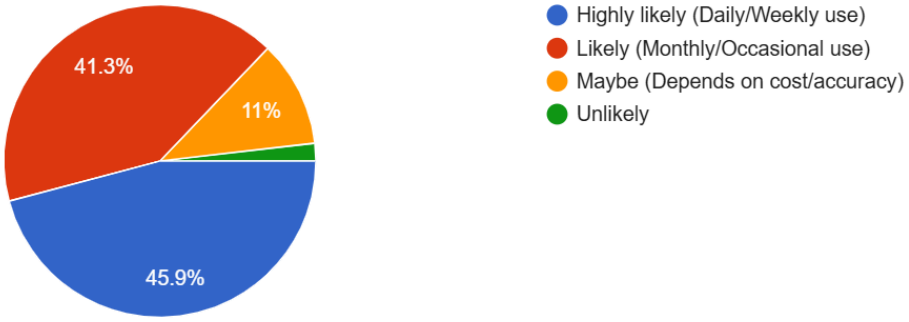
5.How important is it that the tool shows the exact source paragraph (citation) for every answer/summary sentence it generates?

109 responses



6.How likely are you to use an AI-based Research Paper Summarise and Q&A app when it's available?

109 responses



4.5.4 Brainstorming/Observation

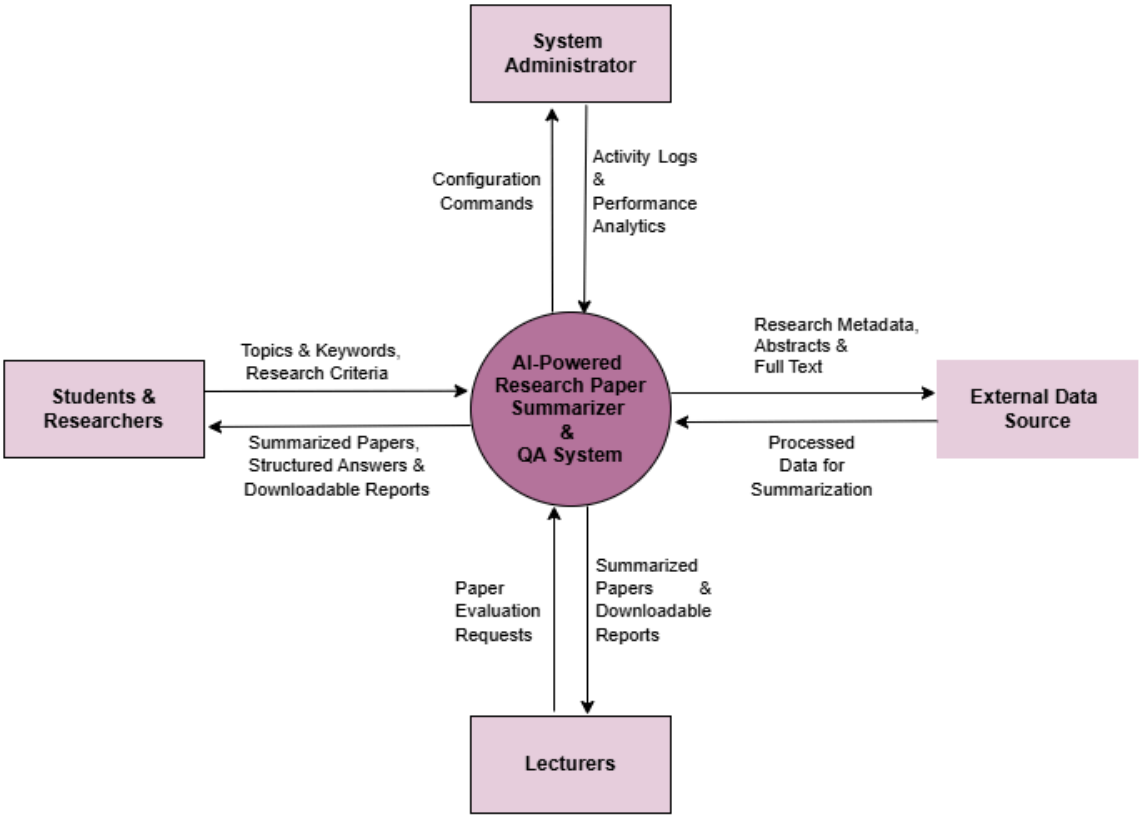
- Integration of BART-base model for high quality abstractive summarization and question answering.
- Inclusion of explainable AI components (e.g., attention visualization) to enhance the transparency of generated summaries and answers.
- Real time summary generation and response display to improve user interaction and research efficiency.
- Support for domain specific terminology handling to ensure the system provides accurate and context aware academic outputs.
- Dashboard design brainstorming emphasized the need for a clean, user friendly interface that allows exporting summaries and reports efficiently.

4.6 Summary of Findings

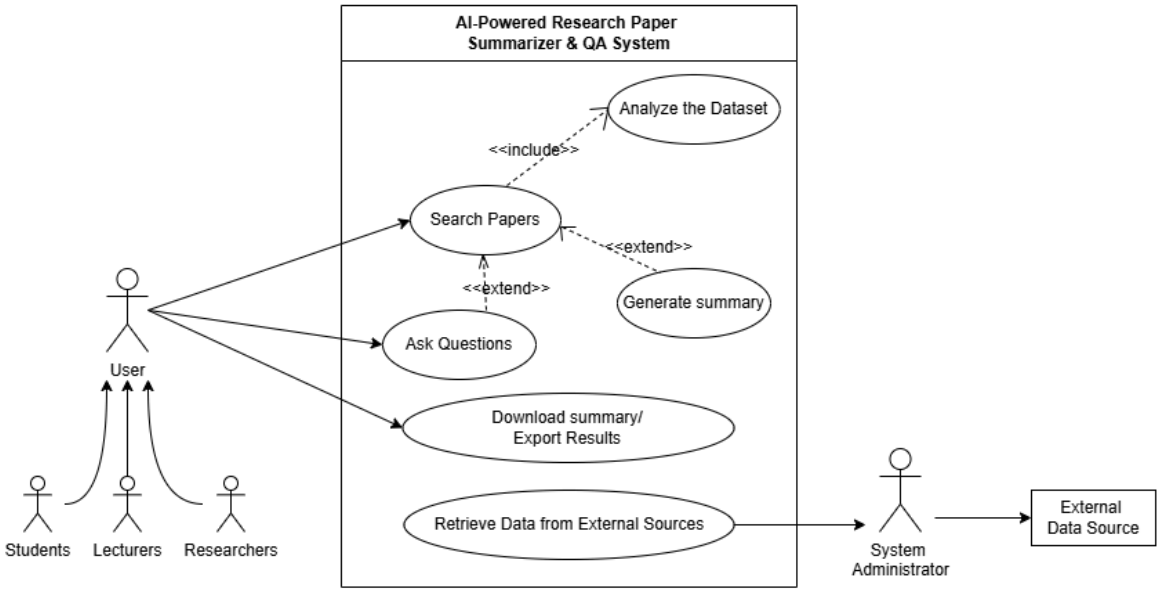
The findings revealed that existing research tools lack integrated summarization and question answering features. Users face challenges in understanding lengthy papers efficiently. The adoption of the BART-base model can enhance content comprehension through accurate, concise summaries and context-based answers. Overall, the system aims to improve research accessibility, reduce reading time, and support academic productivity.

4.7 Context Diagram

The Context Diagram (Level 0 DFD) illustrates how external entities interact with the AI-Based Research Summary and Question Answering System. It provides a high level overview of the system's data flow, showing what inputs are received from each user and what outputs are produced in return.



4.8 Use Case Diagram



4.9 Use Case Descriptions

Use Case ID	Use Case Name	Actor(s)	Description
UC01	Search Papers	Students, Researchers, Lecturers	The user enters a topic or keyword to search for related research papers from external sources (e.g., Google Scholar or Semantic Scholar).
UC02	Generate Summary	Students, Researchers, Lecturers	The user selects a paper, and the system automatically generates a concise summary using NLP techniques.
UC03	Ask Question	Students, Researchers, Lecturers	The user inputs a question (e.g., "What are the limitations?"), and the system provides an answer derived from the paper content.
UC04	Download Summary / Export Results	Students, Researchers, Lecturers	The user can download or export summaries and answers in PDF or text format for offline reference.
UC05	Retrieve Data from External Sources	System Administrator	The system administrator ensures integration with external databases and APIs for automated paper retrieval.
UC06	Manage User Accounts & System Logs	System Administrator	The administrator manages user accounts, monitors system activity, and ensures data security.

4.10 Requirements

4.10.1 Functional Requirements

ID	Requirement	Priority (MoSCoW)
FR01	The system must allow users to search and collect	Must Have

	research papers related to a given topic using web scraping.	
FR02	The system must automatically summarize the main points of each research paper.	Must Have
FR03	The system must allow users to ask questions about the content of the research papers.	Must Have
FR04	The system should generate accurate and concise answers based on the content of the papers.	Should Have
FR05	The system should allow users to download or copy the generated summaries.	Should Have
FR06	The system could store previous searches and summaries for future reference.	Could Have
FR07	The system could support multiple languages for summarization and question answering.	Could Have
FR08	The system will not include plagiarism detection features in the current phase.	Will Not Have

4.10.2 Non-Functional Requirements

ID	**Non-Functional Requirement**	**Description**	**Priority (MoSCoW)**
NFR01	Performance	The system must generate summaries and answers within a reasonable response time (under 5 seconds per query).	Must Have
NFR02	Accuracy	The summarization and question-answering models must achieve at least 85% relevance accuracy in test	Must Have

		evaluations.	
NFR03	Usability	The user interface should be simple, interactive, and easily navigable for both students and researchers.	Should Have
NFR04	Scalability	The system should handle simultaneous requests from multiple users without significant performance loss.	Should Have
NFR05	Security	The system must protect user data and ensure that downloaded papers and summaries are securely processed.	Must Have
NFR06	Maintainability	The system should be designed in a modular way to allow future updates and model replacements.	Could Have
NFR07	Accessibility	The interface could include accessibility options such as font size adjustment and dark mode.	Could Have
NFR08	Reliability	The system must be stable and recover gracefully from unexpected errors.	Must Have

4.11 Chapter Summary

This chapter successfully defined the complete set of requirements for this project. The system architecture was formally bounded by the Context Diagram, and user interactions were detailed in the Use Case Descriptions. The prioritized requirements list, based on the MoSCoW approach, clearly directs the development effort toward achieving essential goals, while non-functional requirements ensure the system is robust and suitable for academic deployment.

Chapter 5: Social, Legal, Ethical and Professional issues (SLEP)

5.1 Chapter overview

The development of AI systems often involves interactions with large volumes of data, including potentially sensitive information. It is essential to adhere to social, legal, ethical, and professional guidelines to ensure responsible development and deployment. This chapter covers the major SLEP concerns encountered during the project, how they were addressed, and the actions taken to comply with the BCS Code of Conduct.

5.2 SLEP issues and mitigation

The following table summarizes the SLEP issues specific to this project and the measures taken to mitigate them.

Social	**Legal**
● **Data Privacy and User Consent:** Social concerns arise from the use of academic papers, which often include personal author information and research data. To mitigate this, we ensure that any paper collected and processed is either publicly available or has the necessary permissions for use. Users providing personal information for feedback or surveys are explicitly asked for their consent to participate. ● **User Accessibility:** The system must be accessible to a wide range of users, including students and early stage researchers. Efforts were made to ensure the system is intuitive and easy to	● **Copyright Concerns:** Legal issues arise regarding the use of copyrighted academic papers. This project ensures that only open access papers, such as those available through arXiv, are used. No copyrighted papers are used without proper licensing or permission. ● **Data Security and Protection:** Legal concerns regarding data security and the protection of user information were addressed by ensuring that the system complies with the General Data Protection Regulation (GDPR) for users within the EU. User data is anonymized and securely stored to prevent unauthorized access.

use, including implementing a simple user interface and providing clear instructions.	
Ethical	**Professional**
● **Bias in Summarization Models:** Ethical concerns regarding bias in AI models were mitigated by using well-established, transparent transformer based models (BART and DistilBERT), which were evaluated for fairness in handling diverse research topics. The model's performance was regularly assessed to ensure no discriminatory patterns were emerging. ● **Accountability in AI Outputs:** Ethical accountability for the AI's output was considered, ensuring that the summarization and question answering systems do not produce harmful, misleading, or factually incorrect information. The project adheres to industry standards by using well-validated, widely accepted AI models and continuously reviewing their outputs for accuracy and reliability.	● **Professional Competence:** Professional issues arise from the need for expertise in AI and NLP to properly design, train, and evaluate the models. The project was supervised by a qualified expert in AI, and constant reviews were conducted to ensure adherence to professional standards in software development. ● **Adherence to Industry Standards:** Professional adherence to the BCS Code of Conduct ensures that the system is developed with high standards of competence, integrity, and respect for the privacy and dignity of users. The system's architecture and its outputs are regularly audited by the development team to maintain professional quality.

5.3 Chapter summary

This chapter addressed the social, legal, ethical and professional issues relevant to this project. It also emphasized the importance of SLEP guidelines to ensure that the system benefits users while adhering to legal and ethical standards.

Chapter 6 : Design

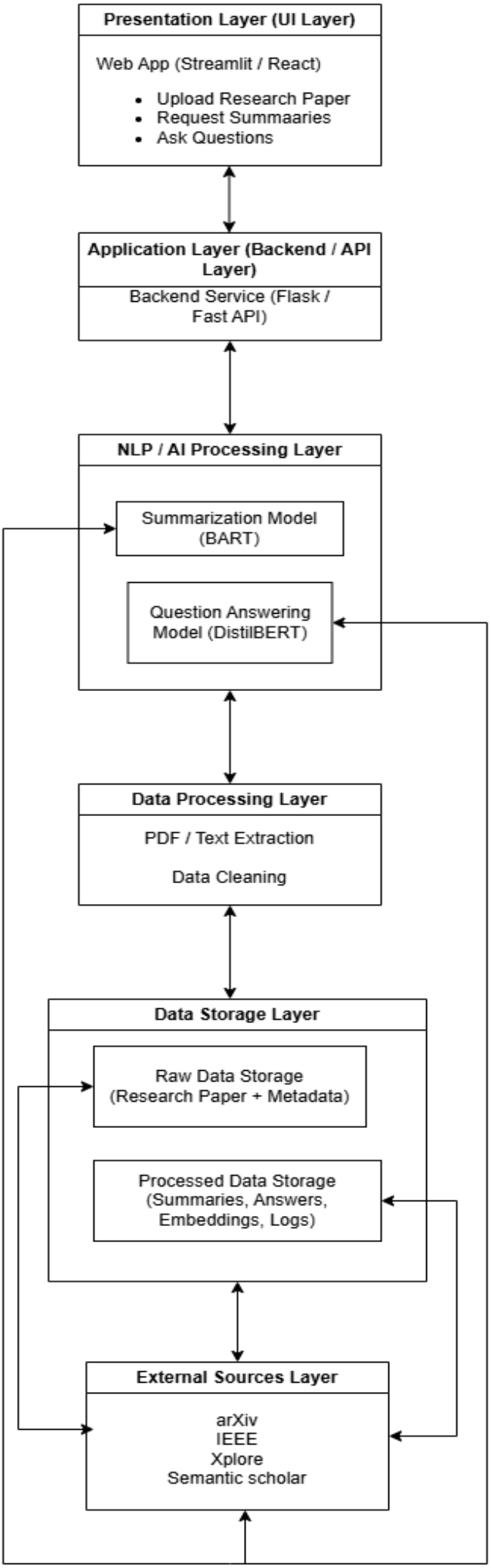
6.1 Chapter Overview

This chapter introduces the design phase of the project, highlighting its importance in turning the project's objectives into a practical plan for building the system. This chapter explains how the design ensures that the AI-based Research Summary and Question Answering System meets its goals, such as providing accurate summaries and supporting user queries. It focuses on key design elements like performance, usability, and accuracy, which are crucial for the system's success. The chapter will cover the design goals, system architecture, detailed design diagrams, and the algorithm behind the AI model. It will also include low fidelity wireframes for the user interface, explaining how the design aims to make the system easy to use. Ultimately, the design phase plays a critical role in shaping the system to be efficient, reliable, and user-friendly, ensuring it meets both functional and non-functional requirements.

6.2 Design Goals

The design goals are essential attributes that define the overall quality of the software system and will be maintained throughout the system. These goals align with the non-functional requirements (NFRs) identified in the System Requirements Specification (SRS) chapter, such as performance, usability, and accuracy. When incorporating these goals into the system design, it is important to think about how each goal will be achieved during both the design and implementation phases. For example, performance will be addressed by optimizing the AI models for speed and efficiency, usability will be ensured through a simple and intuitive user interface, and accuracy will be focused on by carefully selecting and fine-tuning the models to provide reliable summaries and answers. These design goals will guide the development process to ensure that the system meets its intended purpose and provides a seamless experience for the users.

6.3 System Architecture Diagram


Presentation Layer (UI Layer)
Web App (Streamlit / React)
Upload Research Paper
Request Summaaries
Ask Questions
Application Layer (Backend / API Layer)
Backend Service (Flask / Fast API)
NLP / AI Processing Layer
Summarization Model (BART)
Question Answering Model (DistilBERT)
Data Processing Layer
PDF / Text Extraction
Data Cleaning
Data Storage Layer
Raw Data Storage (Research Paper + Metadata)
Processed Data Storage (Summaries, Answers, Embeddings, Logs)
External Sources Layer
arXiv
IEEE
Xplore
Semantic scholar

The high-level architecture diagram of the AI-based Research Summary and Question Answering System illustrates the system's tiered/layered structure. This diagram provides a visual representation of how the different components of the system interact, from the user interface (UI) through to the backend, AI models, data processing, and external sources of research papers.

The system is structured into several distinct layers, each with specific roles:

1. Presentation Layer (UI Layer):

This is the user-facing layer where users interact with the system. The web application (built using Streamlit or React) allows users to upload research papers, request summaries, and ask specific questions about the content. The user input is sent to the backend for processing.

2. Application Layer (Backend/API Layer):

This layer manages the logic of the system. It acts as a bridge between the front-end user interface and the NLP/AI models. The backend, implemented using Flask or FastAPI, handles the user requests, processes them, interacts with the AI models, and returns the results to the presentation layer.

3. NLP/AI Processing Layer:

This layer contains the core Natural Language Processing (NLP) and AI models. The Summarization Model (BART) generates concise summaries of the research papers, while the Question Answering Model (DistilBERT) provides answers to user queries based on the content of the papers. The Semantic Search Model (MiniLM) helps retrieve the most relevant papers based on user input.

4. Data Processing Layer:

This layer is responsible for preparing the raw data (research papers) for the AI models. It includes processes like PDF/Text Extraction (converting papers into readable text) and Data Cleaning (removing irrelevant content like citations, figures, or LaTeX syntax) to ensure the data is in the correct format for model input.

5. Data Storage Layer:

The data storage layer consists of two components:

- **Raw Data Storage:** Stores the original research papers and their metadata (e.g., title, authors, abstract).
- **Processed Data Storage:** Stores the outputs generated by the system, including summaries, answers, and embeddings.

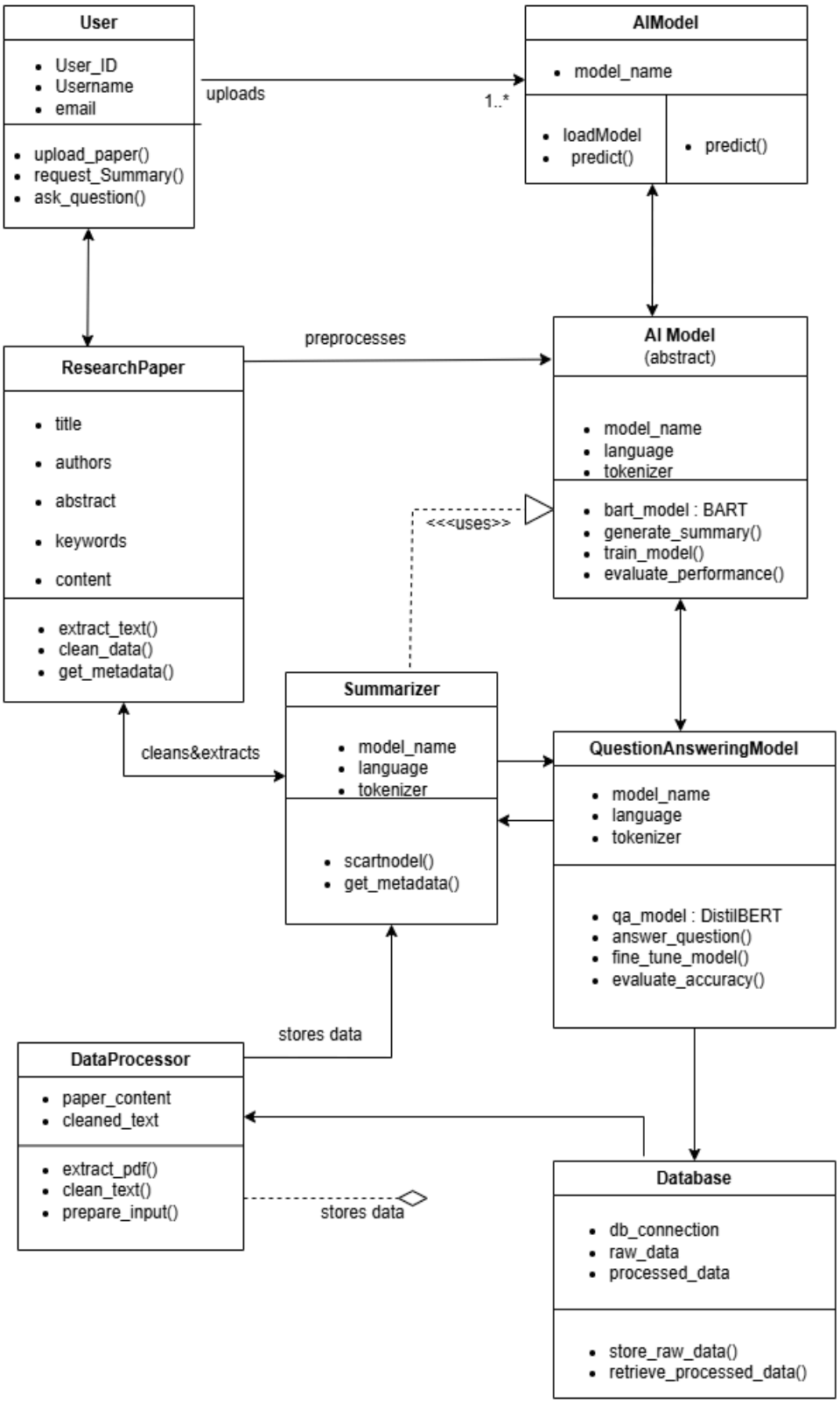
6. External Sources Layer:

This layer represents the sources from which research papers are retrieved. External databases like arXiv, IEEE Xplore, and Semantic Scholar are used to gather research papers for processing. If real time scraping is not implemented, precollected research papers may be used from a local dataset.

6.4 Detailed Design

OOADM (Object-Oriented Analysis and Design Methodology) was chosen for this project because it provides a modular and scalable approach, making it ideal for complex systems like the AI-based Research Summary and Question Answering System. This methodology allows the system's components, such as users, research papers, and AI models, to be represented as objects, each with its own data and behavior. The system's interactions can be easily modeled using sequence diagrams and activity diagrams, ensuring clarity in the data flow and user interaction. OOADM's flexibility supports easy updates and extensions, such as adding new AI models or features. It also aligns with industry practices, ensuring maintainability and scalability, making it the best choice for representing and managing the system's components.

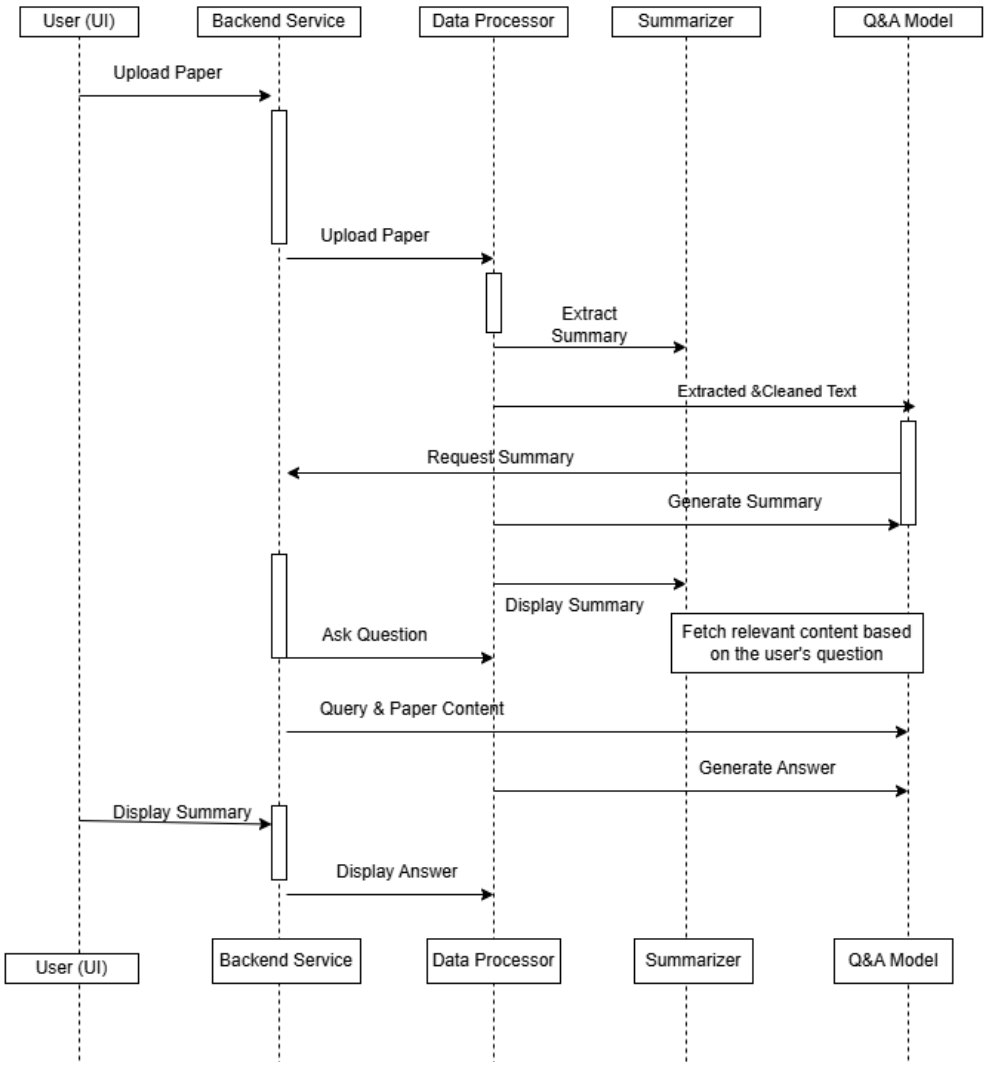
1. Class Diagram



This Class Diagram represents the structure of the AI-based Research Summary and Question Answering System, illustrating the key classes, their attributes, methods, and relationships. The User class allows for uploading research papers, requesting summaries, and asking

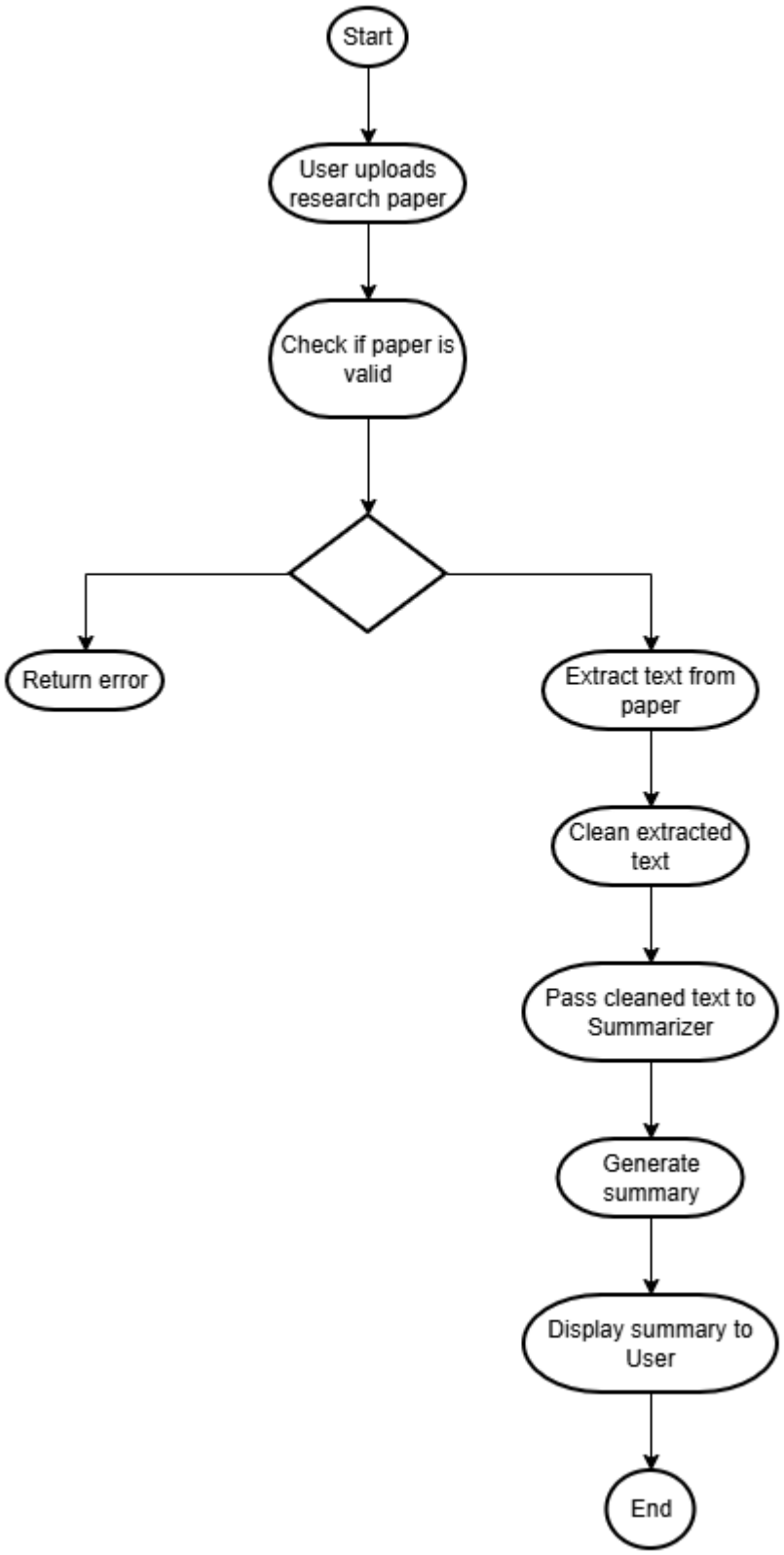
questions. The ResearchPaper class stores metadata and content, with methods for extracting text, cleaning data, and retrieving metadata. The Summarizer and QuestionAnsweringModel classes, both inheriting from the AIModel class, handle the summarization and question-answering tasks, respectively, using models like BART and DistilBERT. The DataProcessor class is responsible for extracting and preparing data from the research papers. Finally, the Database class stores both raw and processed data, including papers, summaries, and answers. Relationships between classes are shown, such as the User uploading a ResearchPaper, which is then processed by the Summarizer and QuestionAnsweringModel, with data stored in the Database.

2. Sequence Diagram



This Sequence Diagram illustrates the flow of operations when a User uploads a research paper, requests a summary, and receives an answer. The process begins with the User uploading the paper through the UI, which is then sent to the Backend Service and processed by the Data Processor. The Data Processor extracts and cleans the paper's content before passing the cleaned text to the Summarizer. The User requests a summary, which triggers the Summarizer to generate it and send it back to the Backend Service, where it is displayed to the User. Afterward, the User asks a question related to the paper, which the Backend Service forwards to the Q&A Model. The Q&A Model fetches relevant content based on the query and generates an answer, which is sent back to the Backend Service and displayed to the User. This diagram demonstrates the sequence of interactions between the UI, Backend Service, Data Processor, Summarizer, and Q&A Model, outlining how the system processes requests and provides responses to the User.

3. Activity Diagram

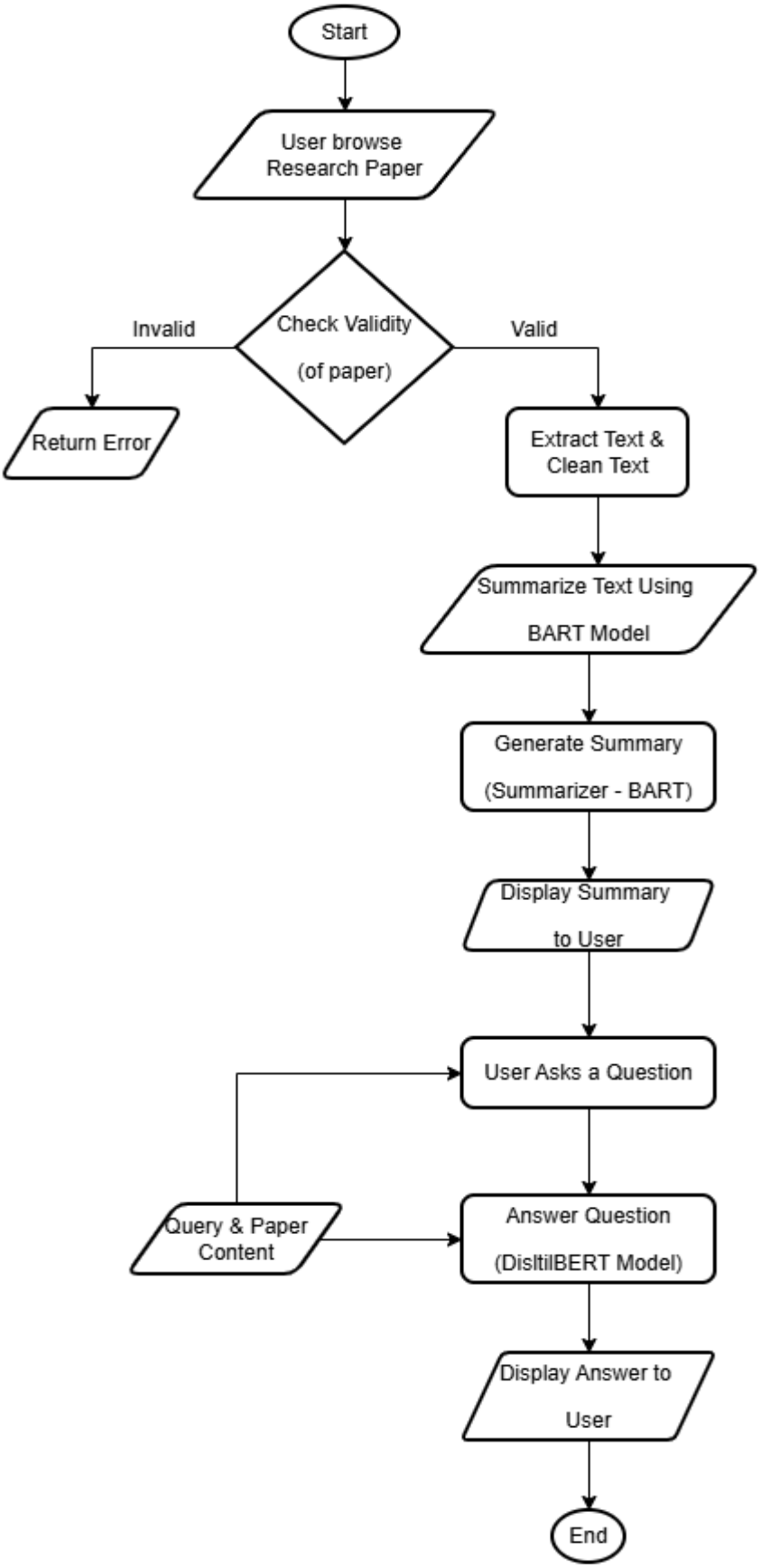


The Activity Diagram illustrates the process of summarizing a research paper in the AI-based Research Summary and Question Answering System. It begins with the user uploading a research paper, after which the system checks if the paper is valid, such as ensuring it's in the

correct format. If the paper is invalid, the process ends with an error message. If valid, the system extracts the text from the paper, cleans it by removing irrelevant content like references, and passes the cleaned text to the Summarizer. The Summarizer generates a concise summary of the paper, which is then displayed to the user. The process concludes with the summary being shown, marking the end of the summarization workflow.

6.5 Algorithm Design

Flow chart

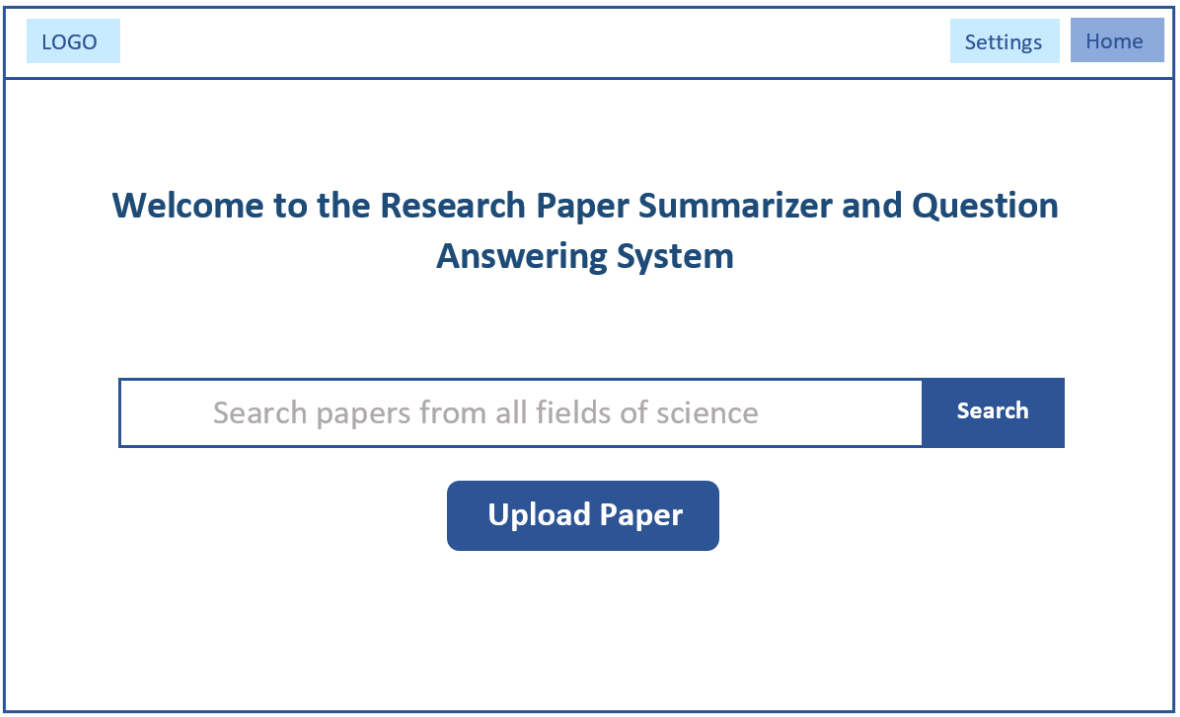

Start
User browse Research Paper
Check Validity (of paper)
Invalid
Valid
Return Error
Extract Text & Clean Text
Summarize Text Using BART Model
Generate Summary (Summarizer - BART)
Display Summary to User
User Asks a Question
Query & Paper Content
Answer Question (DisltilBERT Model)
Display Answer to User
End

The Algorithm Design Flowchart outlines the process flow of the AI-based Research Summary and Question Answering System. It begins when the User browses a research paper to the system. The system then checks if the paper is valid, and if it's not, an error message is returned. If the paper is valid, the system extracts and cleans the text, removing unnecessary elements like references and figures. The cleaned text is passed to the Summarizer (BART model), which generates a concise summary of the paper. This summary is then displayed to the User. The User can also ask a question related to the research paper, which is processed by the DistilBERT model to generate an answer. Finally, the system displays the answer to the User, completing the process.

6.6 UI Design

Wireframes are visual representations of the interface. Below are the low fidelity wireframes for the various screens of the AI-based Research Summary and Question-Answering System.

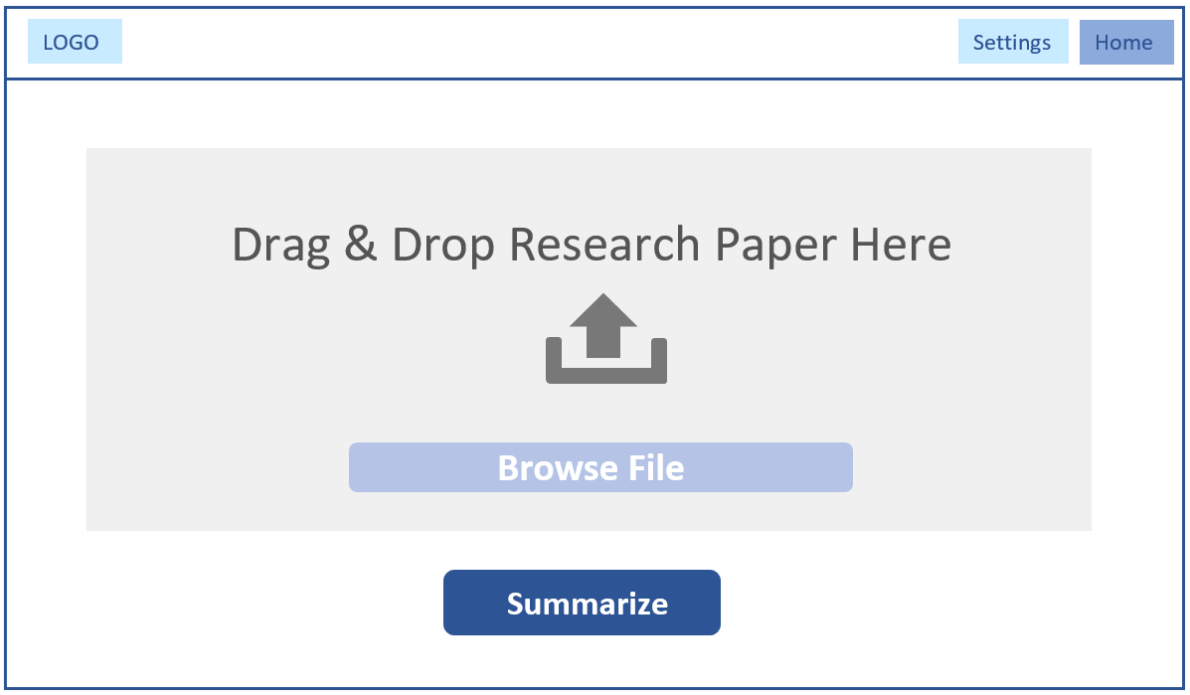
Home Page



This is the landing page where the user can either search for papers or upload a research paper.

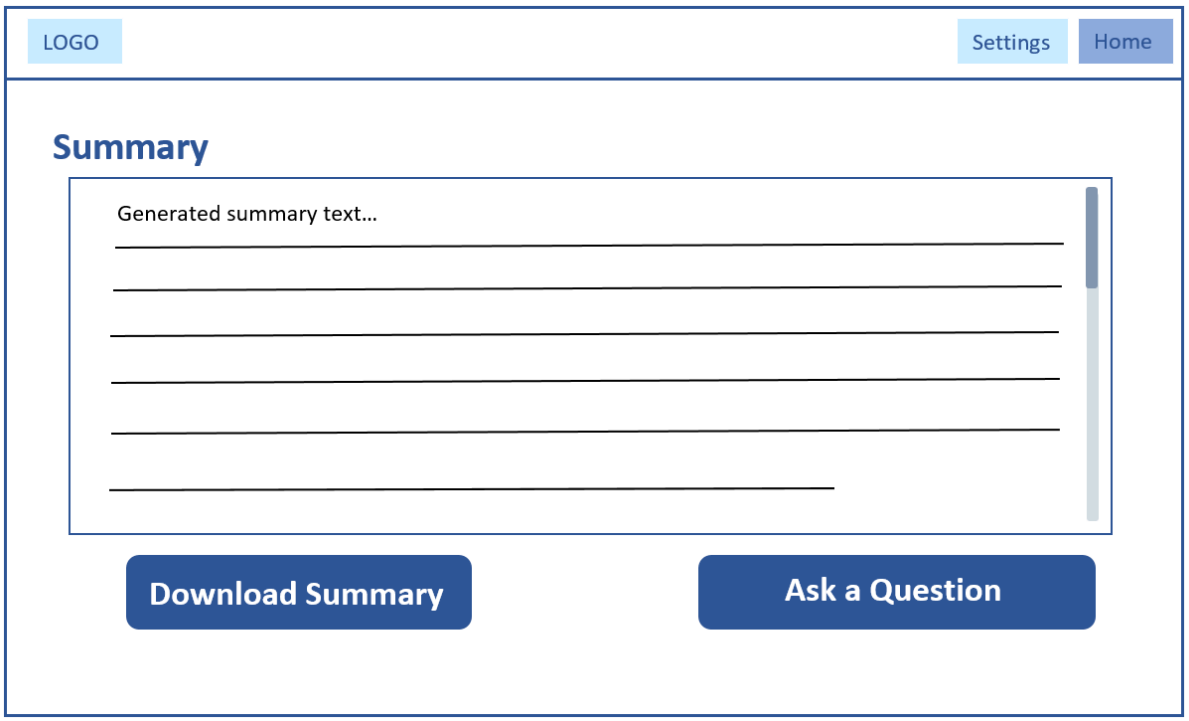
Upload Research Paper Page

The user can drag and drop or browse to upload the research paper for summarization.



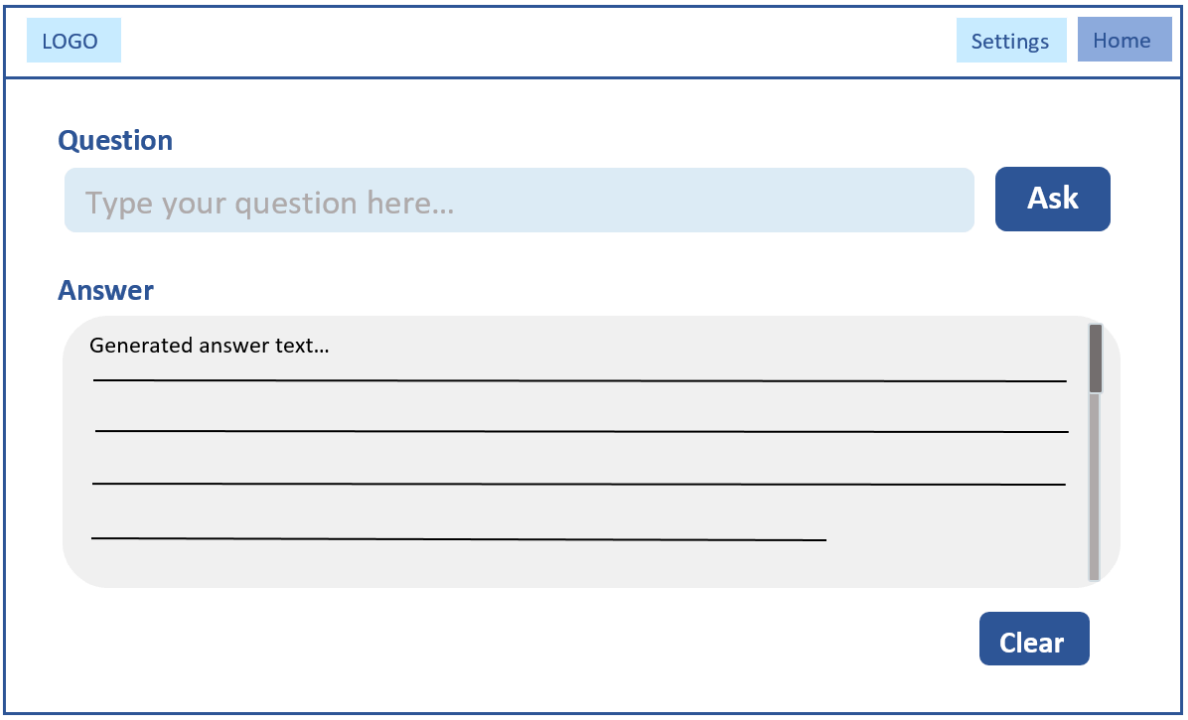
Summary Display

This screen displays the generated summary of the uploaded research paper.



Question and Answer Screen

In this screen, the user can ask a question related to the research paper and receive an answer.



6.7 Chapter Summary

In this chapter, we explored the design phase of the AI-based Research Summary and Question Answering System. We discussed the design goals, which include ensuring the system is efficient, accurate, and user-friendly. The system's architecture was broken down into different layers, from the user interface to the backend and AI models. We also reviewed the detailed design, including class diagrams, sequence diagrams, and activity diagrams, to illustrate the system's structure and flow. Additionally, the algorithm for summarizing research papers and answering user queries was presented, showing how AI models like BART and DistilBERT are used. Finally, we focused on the UI design, highlighting wireframes that represent the system's interface and user interactions, ensuring that the system is both accessible and easy to use for all users.

Chapter 7: Implementation

7.1 Chapter Overview

The Implementation Phase is a critical stage in the development of the AI-based Research Summary and Question Answering System, where the designs and plans created in the previous chapter are transformed into functional software components. This phase brings the system to life by translating the system's architecture, algorithms, and UI designs into actual code and working features.

The main objectives of this phase are to:

- **Develop Core Functionalities:** Implement the algorithms, data processing, and AI models, such as the BART summarization model and DistilBERT question answering model.
- **Integrate Frontend and Backend:** Ensure that the user interface (UI) components interact seamlessly with the backend services to deliver a smooth user experience.
- **Ensure System Usability and Accessibility:** Implement UI features that are user-friendly and accessible to all user groups.
- **Ensure Performance and Scalability:** Address performance issues by optimizing code and using appropriate tools and frameworks to handle the anticipated workload efficiently.

This chapter will provide a detailed exploration of the technologies used, how the system's core functionalities were implemented, and the integration of the various modules. It will also highlight the challenges faced during the implementation process and the solutions devised to overcome them. The chapter will end with a summary of the key aspects of the implementation and a transition to the testing phase.

The Implementation Chapter directly builds upon the Design Chapter, where we laid the groundwork for how the system should function. The designs, algorithms, and system structure presented previously are now being translated into working software components, which will be thoroughly tested in the subsequent chapters.

7.2 Technology Selection

The Technology Selection section outlines the tools, frameworks, and programming languages used in the development of the AI-based Research Summary and Question Answering System. This selection was made based on the system requirements, the need for AI/ML model support, and ensuring that the solution is both efficient and scalable.

7.2.1 Technology Stack

The Technology Stack for this project includes the following layers:

1. **Frontend Layer:**
    - **React.js:** Chosen for building the user interface (UI). React is highly efficient, flexible, and enables the creation of dynamic web pages.
2. **Backend Layer:**
    - **Flask:** A lightweight Python web framework for building the backend services that manage communication between the frontend and the AI models.
3. **AI/ML Layer:**
    - **PyTorch:** The primary deep learning framework used to build and train the BART (summarization model) and DistilBERT (question answering model). PyTorch is preferred for its ease of use and dynamic computation graphs, ideal for model training and deployment.
    - **Hugging Face Transformers:** A library built on top of PyTorch to provide pretrained models such as BART and DistilBERT, which are used for summarization and question answering tasks.
4. **Data Layer:**
    - **MongoDB:** A NoSQL database used to store research papers, metadata, generated summaries, and user queries.
5. **Cloud Hosting/Deployment:**
    - **AWS EC2:** Cloud servers for hosting the application and running AI model inference tasks.

7.2.2 Programming Languages

- **Python:** Python was chosen as the main programming language due to its rich ecosystem for machine learning and data science. Libraries like PyTorch, Hugging Face Transformers, Flask, and Pandas provide powerful tools for AI model development, data processing, and web development.
- **JavaScript:** JavaScript (with React.js) was chosen for the frontend development to create a responsive and interactive user interface.

7.2.3 Development Framework

- **Flask:** Flask was selected as the backend framework due to its simplicity and flexibility. It allows easy integration with machine learning models, handles HTTP requests, and serves the AI model responses to the frontend interface. Flask's lightweight nature makes it ideal for this project where we need a responsive and scalable backend.
- **React.js:** React provides a declarative and component based approach to building the UI, making it easier to create a dynamic and user-friendly interface for research paper uploads, summaries, and Q&A features.

7.2.4 Libraries/ Toolkits

- **PyTorch:** PyTorch was selected for training and running the BART and DistilBERT models, as it provides a highly flexible deep learning framework that supports dynamic graphs and GPU acceleration. This is essential for AI tasks like summarization and question answering.
- **Hugging Face Transformers:** The Transformers library provides access to pretrained models like BART for summarization and DistilBERT for question answering, which can be finetuned or used directly for efficient model deployment.
- **Pandas and NumPy:** These libraries are used for data manipulation, cleaning, and preprocessing, helping with the extraction and cleaning of research paper data.
- **Flask-RESTful:** Flask-RESTful simplifies API development for the backend, allowing the frontend to easily interact with the backend services.
- **MongoDB:** A NoSQL database used for storing raw research papers, their summaries, and user queries. It is scalable and flexible, making it a good fit for storing large amounts of unstructured data.

7.2.5 Integrated Development Environment (IDEs)

- **Visual Studio Code (VS Code):** VS Code was used as the primary IDE due to its excellent support for both Python and JavaScript. It has built in features like IntelliSense, code debugging, and Git integration, which streamline the development process.
- **Jupyter Notebook:** Jupyter Notebooks were used during the model development phase for experimentation, data exploration, and testing machine learning models.

7.2.6 Summary of the Technology Selection

The technology stack was carefully selected to meet the functional and non-functional requirements outlined in the System Requirements Specification (SRS) chapter. Python was chosen for backend and AI model implementation due to its rich ecosystem in machine learning and data science, while Flask serves as the backend framework. React.js was chosen for the frontend to build an interactive and responsive user interface. The PyTorch and Hugging Face Transformers libraries provide powerful pretrained models for summarization and question answering. MongoDB is used for flexible and scalable storage of research papers and generated summaries. Together, this technology stack ensures that the system is efficient, scalable, and easy to maintain.

7.3 Core Functionalities Implementation

The Core Functionalities Implementation section details the development and integration of the key features that make up the AI-based Research Summary and Question Answering System. This includes explaining how data is processed, how AI models are trained and utilized for summarization and question answering, the modular development approach, the logical organization of the code, and the integration of various components.

Dataset and Split Sizes

- **Dataset:** The arXiv Scientific Research Papers Dataset was used for this project, consisting of research papers with various categories.
- **Data Split:**
    - **Training Data:** 70% of the dataset was used to train the AI models (e.g., BART for summarization and DistilBERT for question answering).

- **Validation Data:** 15% of the dataset was used for validating the model during training to optimize hyperparameters and avoid overfitting.
- **Testing Data:** 15% of the dataset was reserved for testing the model's performance after training.

This split was performed using stratified sampling, ensuring that the distribution of categories was maintained across all splits, providing balanced training, validation, and test sets.

Code for data split:

```
# Split the data
train_df, val_df, test_df = DataSplitter.stratified_split(
    df_filtered,
    category_column='category_code',
    train_ratio=0.7,
    val_ratio=0.15,
    test_ratio=0.15,
    random_state=42
)
```

Modular Development

The system is developed in a modular fashion to maintain flexibility, scalability, and ease of maintenance. Each module focuses on a specific functionality and communicates with others through well-defined interfaces.

1. Data Preprocessing

Purpose: The data preprocessing module is responsible for extracting, cleaning, and preparing research papers for input to the AI models. This step is essential to ensure that the data is in the right format and that irrelevant information is removed.

Implementation:

- **Text Extraction:** The raw research papers, typically in PDF format, are first converted into readable text using PDF parsers like pdfminer. This allows the system to extract the full text of the paper, including the abstract, introduction, conclusion, and keywords.
- **Text Cleaning:** After extracting the text, the next step is to clean the data by removing unnecessary content such as:
    - Citations and references
    - Figures, tables, or LaTeX code
    - Special characters or unwanted symbols

- Stopwords (if applicable, although this can be controlled via preprocessing settings)

- **Tokenization:** The cleaned text is tokenized into words or subwords, depending on the model requirements (e.g., BART and DistilBERT use subword tokenization). Tokenization breaks the text into smaller units, making it easier for the models to process.

Data preprocessing code:

```
import pandas as pd
import os
from preprocess import DataPreprocessor, DataSplitter

# Check if the raw data exists and load the dataset
raw_data_path = 'arXiv Scientific Research Papers Dataset (2).csv'

if os.path.exists(raw_data_path):
    # Load raw data
    print(f"Loading raw dataset from {raw_data_path}...")
    df = pd.read_csv(raw_data_path)
    print(f"Raw dataset loaded: {len(df)} samples")

    # Process the data
    preprocessor = DataPreprocessor(remove_stopwords=False)
    df_processed = preprocessor.preprocess_dataframe(df, text_column='summary')
```

Relating to Functional Requirements:

This preprocessing step ensures that the input text is ready for the summarization and question answering models, which are essential functions described in the SRS. Clean data is crucial for the models to generate accurate and meaningful outputs.

2. Model Training

Purpose: The Model Training module involves training the BART (for summarization) and DistilBERT (for question answering) models on the dataset to generate accurate summaries and answers.

Implementation:

- **BART Model for Summarization:**
  - The BART model is a pretrained sequence-to-sequence model optimized for text generation tasks like summarization.
  - The model was fine-tuned on the dataset of research papers (after preprocessing) to generate summaries that capture the most relevant information from the paper.

- Fine-tuning was performed using PyTorch and Hugging Face's Transformers library.
- The model was trained with a batch size of 16, using the Adam optimizer with a learning rate of 5e-5, and evaluated using ROUGE metrics for summary quality.

```
from transformers import BartForConditionalGeneration, BartTokenizer

# Load pre-trained BART model and tokenizer
model = BartForConditionalGeneration.from_pretrained('facebook/bart-large-cnn')
tokenizer = BartTokenizer.from_pretrained('facebook/bart-large-cnn')

# Tokenize the training data
inputs = tokenizer(df_processed['summary'].tolist(), return_tensors='pt', padding=True, truncation=True)

# Training configuration (e.g., learning rate, optimizer, etc.)
```

- **DistilBERT Model for Question Answering:**
  - DistilBERT is a smaller, faster version of BERT, optimized for tasks like question answering. It was trained on a SQuAD-like dataset of research papers and summaries.
  - The model was fine-tuned using PyTorch and the Hugging Face Transformers library, focusing on the task of finding the answer to a given question based on the content of the research paper.
  - Loss functions like cross-entropy loss were used to optimize the model's performance during training.

```
from transformers import DistilBertForQuestionAnswering, DistilBertTokenizer

# Load pre-trained DistilBERT model and tokenizer
model = DistilBertForQuestionAnswering.from_pretrained('distilbert-base-uncased-distilled-squad')
tokenizer = DistilBertTokenizer.from_pretrained('distilbert-base-uncased-distilled-squad')

# Tokenize the context and question
inputs = tokenizer(question, context, return_tensors='pt')
```

Relating to Functional Requirements:

- **Summarization:** The BART model directly addresses the requirement to generate summaries of research papers.

- **Question Answering:** The DistilBERT model meets the requirement of answering questions related to the content of the papers.

3. Prediction Engine

Purpose: The Prediction Engine is responsible for generating summaries and answering queries using the trained models.

Implementation:

- **Summarization Prediction:**
  - After training the BART model, the system can generate summaries by passing the cleaned and tokenized text through the model.
  - The generate_summary() method in the Summarizer class is called to provide the final summary output.

```
# Function to generate summary using BART
def generate_summary(text):
    inputs = tokenizer(text, return_tensors='pt', max_length=1024, truncation=True)
    summary_ids = model.generate(inputs['input_ids'], max_length=150, num_beams=4, early_stopping=True)
    summary = tokenizer.decode(summary_ids[0], skip_special_tokens=True)
    return summary
```

- **Question Answering Prediction:**

  For answering user queries, the system uses the DistilBERT model. The answer_question() method of the QuestionAnsweringModel class processes the user's query and the research paper summary, returning the relevant answer.

```
# Function to answer questions using DistilBERT
def answer_question(question, context):
    inputs = tokenizer(question, context, return_tensors='pt')
    outputs = model(**inputs)
    start_scores = outputs.start_logits
    end_scores = outputs.end_logits

    # Find the tokens with the best start and end score
    start_index = torch.argmax(start_scores)
    end_index = torch.argmax(end_scores)

    # Decode the answer from tokens
    answer_tokens = inputs.input_ids[0][start_index:end_index+1]
    answer = tokenizer.decode(answer_tokens)
    return answer
```

- **Example of Prediction Workflow:**

1. **User Uploads Paper:** The user uploads a research paper, and the system preprocesses the text.

2. **Summarization:** The preprocessed paper is passed to the BART model to generate the summary.
3. **Question Answering:** The user asks a question, which is passed to the DistilBERT model along with the paper content or summary to generate an answer.

Relating to Functional Requirements:

- **Generating Summaries:** This functionality is linked to the summarization feature in the SRS.
- **Answering Queries:** This is directly related to the question answering requirement, which allows users to extract specific information from the research papers.

These modules (data preprocessing, model training, and the prediction engine) work together to fulfill the functional requirements defined in the SRS. The data preprocessing ensures that the input data is clean and ready for the models, model training ensures that the models are fine-tuned to generate accurate outputs, and the prediction engine seamlessly integrates the trained models to generate summaries and answer user queries. These modules are implemented using industry-standard tools like PyTorch, Hugging Face Transformers, and Flask to ensure a robust and efficient system.

7.4 User Interface Implementation

The User Interface (UI) of the AI-based Research Summary and Question-Answering System was designed with the goal of providing a user-friendly and accessible experience. This section details the frontend development, backend integration, and accessibility considerations that were implemented to ensure the system meets the needs of all users.

Frontend Development

The frontend of the system was built using React.js to create an interactive, responsive, and easy-to-navigate interface. The key UI components implemented include:

1. **Home Page:** The homepage includes a file upload form where the user can upload a research paper. Additionally, the user can interact with the system via a search bar to search for research papers. This page serves as the starting point for all user interactions.

2. **Upload Research Paper Page:** This page provides a drag-and-drop area or browse file button, allowing users to upload their research papers for summarization. Once uploaded, the system processes the paper in the backend.
3. **Summary Display:** After the paper is uploaded and processed, the Summary Display page shows the generated summary. The user can download the summary or proceed to ask questions based on the content.
4. **Q&A Screen:** On the Q&A screen, the user can input a question related to the research paper. The question is processed in the backend using the DistilBERT model, and the answer is displayed to the user.

These components were designed with usability in mind, ensuring that the system is easy to navigate, and the user can easily interact with the system at each step.

Backend Integration

The frontend communicates with the backend using API calls, allowing seamless interaction between the user interface and the model inference. The system is built using Flask, which handles incoming requests from the frontend and routes them to the appropriate model for processing. Key interactions include:

- **Upload Paper:** When the user uploads a research paper, the UI sends the file to the backend, where it is preprocessed and passed to the BART model for summarization.
- **Generate Summary:** The backend processes the uploaded research paper and calls the BART model to generate a summary. The result is sent back to the UI, which displays it to the user.
- **Ask Question:** When a user asks a question, the UI sends the query to the backend, which processes it using the DistilBERT model for question answering. The answer is then returned to the UI for display.

The Flask API ensures smooth communication between the frontend and backend, allowing the user to interact with the system without any delays.

Accessibility Considerations

Ensuring that the system is accessible to a diverse range of users is a key focus of the UI design. The following accessibility features were incorporated into the system:

1. **Keyboard Navigation:** All interactive elements, such as buttons and forms, can be accessed using the keyboard alone. This ensures that users with motor impairments can navigate the system without needing a mouse.

2. **Screen Reader Support:** Alt text was added to all images and buttons for users who rely on screen readers. This ensures that the system is usable by visually impaired users.
3. **Color Contrast and Readability:** The UI uses high color contrast between text and background to improve readability for users with color blindness or visual impairments. Additionally, users can adjust the font size for better readability.
4. **Responsive Design:** The UI was built to be responsive, ensuring it works well on both desktop and mobile devices. This ensures that users with different devices can easily access and use the system.
5. **Error Handling:** Clear and informative error messages were implemented to guide users when something goes wrong, such as when an invalid file is uploaded or a query fails.

The User Interface Implementation focuses on providing a seamless and accessible experience for users. The frontend development ensures that users can easily interact with the system, while backend integration handles the necessary processing and model inference. Accessibility considerations were taken into account to ensure the system is usable by a wide range of users, including those with disabilities.

7.5 Challenges and Solutions

During the implementation of the AI-based Research Summary and Question Answering System, several challenges were encountered. These challenges were related to debugging, resource constraints, and performance bottlenecks. Below is a discussion of these challenges and the solutions implemented to address them.

1. Debugging and Model Fine-Tuning

Challenge: One of the significant challenges was debugging the AI models, specifically during the fine-tuning process of the BART and DistilBERT models. When training the models on the research paper dataset, we faced issues with the models producing summaries or answers that were inconsistent or irrelevant.

Solution:

- The issue was addressed by carefully monitoring training using metrics like loss and accuracy. By adjusting the learning rate, increasing the batch size, and using different optimizer configurations, the model's performance improved.

- Additionally, fine-tuning the models using the Hugging Face Transformers library allowed us to experiment with various configurations to optimize performance.
- Validation sets were used to tune hyperparameters and prevent overfitting, ensuring that the model would generalize well to unseen data.

2. Resource Constraints and Model Inference Speed

Challenge: Running large-scale models like BART and DistilBERT posed challenges in terms of computational resources. Training these models required significant GPU power and memory, and inference (especially for summarization) was relatively slow.

Solution:

- To address this, we used GPU acceleration during training and inference, leveraging cloud-based GPU instances (e.g., AWS EC2 or Google Colab) to speed up model processing.
- During inference, we also optimized the model's batch processing to handle multiple inputs simultaneously, reducing the overall time taken for summarization and question answering.
- Additionally, we utilized model pruning and quantization techniques to reduce the model size, which led to faster predictions without compromising accuracy significantly.

3. Performance Bottlenecks in Data Processing

Challenge: Another challenge was handling the large dataset of research papers. Text extraction from PDFs and processing the content for the summarization model was slow, especially when the number of documents increased. This created performance bottlenecks, particularly when dealing with multiple users uploading papers simultaneously.

Solution:

- Parallel Processing was implemented using Python’s multiprocessing library to speed up the text extraction and data cleaning steps.
- Distributed computing was used for large-scale data processing, where multiple tasks were executed in parallel to improve throughput.
- We also used efficient data structures (like Pandas DataFrames) and caching to store intermediate results and avoid redundant computations.

4. Database Query Optimization

Challenge: As the system grew and more papers were uploaded, we encountered performance issues with the database. Slow database queries affected the user experience, particularly when retrieving papers or summary data. The system's performance degraded as the amount of stored data increased.

Solution:

- Indexing was added to frequently queried fields in the database (e.g., paper titles, categories), improving the speed of read operations.
- Database queries were optimized by reducing the number of joins and minimizing data retrieval to only necessary fields. We also implemented pagination for large result sets to avoid overwhelming the system.
- We switched to using a NoSQL database (MongoDB) for faster data retrieval, especially when dealing with unstructured data like research papers and summaries.

5. Integration and System Coordination

Challenge: Integrating the frontend and backend systems posed challenges, particularly in terms of ensuring seamless communication between the UI and AI models. API calls had to be correctly structured to allow for efficient data flow, and synchronization between the frontend and backend was sometimes problematic.

Solution:

- We implemented asynchronous API calls to improve communication between the frontend and backend, ensuring that the UI remains responsive even when processing large papers or multiple queries.
- To handle potential timeouts and improve overall responsiveness, we used task queues (such as Celery) to manage long-running tasks in the backend, allowing for a more responsive user interface.
- Detailed error handling and logging mechanisms were put in place to debug issues and ensure that failed requests were handled gracefully without impacting the user experience.

Throughout the implementation of the AI-based Research Summary and Question Answering System, we encountered several challenges related to model training, resource constraints, performance bottlenecks, and database optimization. These challenges were addressed by leveraging cloud resources, optimizing code and queries, and using efficient data processing techniques. The solutions implemented ensured that the system is not only functional but also scalable and responsive, providing a smooth user experience even as the system grows.

7.6 Chapter Summary

In this chapter, we covered the Implementation Phase of the AI-based Research Summary and Question Answering System. This phase involved translating the design and planned architecture into functional components, ensuring that the system meets its objectives and performs effectively.

Technologies Used:

We utilized a range of technologies to develop the system, including:

- **Python** for the backend and AI model implementation.
- **React.js** for the frontend to build a dynamic and interactive user interface.
- **Flask** for backend integration, ensuring smooth communication between the frontend and AI models.
- **PyTorch** and **Hugging Face Transformers** to fine-tune pre-trained models like **BART** for summarization and **DistilBERT** for question answering.

Major Modules Implemented:

- **Data Preprocessing:** A module to clean and tokenize text from research papers to prepare it for model input.
- **Model Training:** We fine-tuned BART for summarization and DistilBERT for question answering using preprocessed data.
- **Prediction Engine:** The trained models were integrated to generate summaries and answer user queries.
- **Frontend Development:** The UI was designed to allow users to easily upload research papers, view summaries, and interact with the system to ask questions.

Challenges Overcome:

During implementation, we faced several challenges, including debugging issues with AI models, resource constraints in model inference, and performance bottlenecks in data processing and database queries. These were addressed using cloud resources for faster processing, optimizing the models and queries, and implementing parallel processing for better performance.

With the major modules implemented and challenges addressed, the system is now ready for the Testing Phase. The next step will involve evaluating the system's performance, accuracy, and usability to ensure it meets the requirements outlined in the System Requirements Specification (SRS) chapter.

Chapter 8 : Time Schedule

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